

Theory of Algebraic Topology 5

July 29, 2026

Lecture notes with worked examples in algebraic topology

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- Homology of the projective plane

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- Quotient constructions often encode global topology through local identifications.
- Covering spaces connect topology with algebra through fundamental groups.
- Simplicial approximation turns continuous questions into combinatorial ones.
- Homology measures cycles modulo boundaries.
- Top homology detects orientability.
- Elementary collapses preserve homology and often simplify complexes drastically.
- Products of polyhedra remain polyhedral after triangulation.

1 Problem 6 Homology of the triangulated torus, Klein bottle, and projective plane

1.1 Definitions and background

Euler characteristic and triangulations. In Question 2, explicit triangulations were given for the torus T , the Klein bottle K , and the projective plane $\mathbb{RP}^2$. The previous two questions describe how homology changes when one deletes:

- a maximal simplex, and
- a free face together with the unique simplex containing it.

The second operation is an elementary collapse and does not change homology; the first changes homology only in two adjacent degrees, in a completely controlled way.

Strategy. Using repeated applications of those two results, one may simplify the given triangulations to the standard polygonal models for the torus, Klein bottle, and projective plane. From there, one reads off the usual cellular chain complexes and computes homology.

1.2 Problem statement

Problem. Using the two previous questions, compute the homology groups of the simplicial complexes described in Q2, and describe generators for each homology group.

1.3 Solution

The simplicial complexes from Q2 triangulate the following spaces:

$$T, \quad K, \quad \mathbb{RP}^2.$$

By repeated elementary collapses and deletions of maximal simplices, as described in the previous two questions, each triangulation may be reduced to the usual quotient-square model of the corresponding surface. Therefore the homology groups are the standard homology groups of the torus, Klein bottle, and projective plane.

We compute each in turn.

1. The torus.

The torus has the standard CW structure with:

- one 0-cell,
- two 1-cells a, b ,
- one 2-cell attached along the loop

$$aba^{-1}b^{-1}.$$

Hence the cellular chain complex is

$$0 \longrightarrow C_2(T) \cong \mathbb{Z} \xrightarrow{d_2} C_1(T) \cong \mathbb{Z}^2 \xrightarrow{d_1} C_0(T) \cong \mathbb{Z} \longrightarrow 0.$$

Since there is only one 0-cell, we have

$$d_1 = 0.$$

Also, in the attaching word $aba^{-1}b^{-1}$, the total exponent of both a and b is 0, so

$$d_2 = 0.$$

Therefore

$$\begin{aligned} H_2(T) &\cong \ker d_2 \cong \mathbb{Z}, \\ H_1(T) &\cong \ker d_1 / \operatorname{im} d_2 \cong \mathbb{Z}^2, \end{aligned}$$

and

$$H_0(T) \cong \mathbb{Z}.$$

All higher homology groups vanish:

$$H_j(T) = 0 \quad (j \geq 3).$$

Generators for the torus.

- $H_0(T) \cong \mathbb{Z}$ is generated by the class of any vertex.
- $H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z}$ is generated by the two basic loops:

$$[a], \quad [b],$$

where a is the horizontal loop and b is the vertical loop in the quotient-square model.

- $H_2(T) \cong \mathbb{Z}$ is generated by the fundamental class

$$[T],$$

which may be represented simplicially by the sum of all oriented 2-simplices in the triangulation, with compatible orientations.

Thus

$$H_0(T) \cong \mathbb{Z}, \quad H_1(T) \cong \mathbb{Z}^2, \quad H_2(T) \cong \mathbb{Z}.$$

2. The Klein bottle.

The Klein bottle has the standard CW structure with:

- one 0-cell,
- two 1-cells a, b ,
- one 2-cell attached along the loop

$$aba b^{-1},$$

equivalently $aba^{-1}b$, depending on conventions.

With the usual convention, the boundary map in degree 2 is

$$d_2(1) = 2a,$$

up to a choice of basis in $C_1(K) \cong \mathbb{Z}^2$. Hence the cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

with

$$d_2(1) = (2, 0)$$

after choosing a suitable basis of C_1 .

Therefore

$$H_2(K) = \ker d_2 = 0,$$

and

$$H_1(K) \cong \mathbb{Z}^2 / \langle (2, 0) \rangle \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

Also

$$H_0(K) \cong \mathbb{Z},$$

and

$$H_j(K) = 0 \quad (j \geq 3).$$

Generators for the Klein bottle.

- $H_0(K) \cong \mathbb{Z}$ is generated by the class of any vertex.
- $H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2$ is generated by:
 - one class $[b]$ of infinite order, represented by the untwisted direction in the square model;
 - one class $[a]$ of order 2, represented by the twisted direction.

Thus

$$2[a] = 0, \quad [b] \text{ has infinite order.}$$

- Since the Klein bottle is nonorientable,

$$H_2(K) = 0.$$

There is no global compatible choice of orientations on all 2-simplices producing a nonzero 2-cycle.

So

$$H_0(K) \cong \mathbb{Z}, \quad H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2, \quad H_2(K) = 0.$$

3. The projective plane.

The projective plane has the standard CW structure with:

- one 0-cell,
- one 1-cell a ,
- one 2-cell attached by the map

$$a^2.$$

So the cellular chain complex is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

Therefore

$$\begin{aligned} H_2(\mathbb{RP}^2) &= 0, \\ H_1(\mathbb{RP}^2) &\cong \mathbb{Z}/2, \end{aligned}$$

and

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}.$$

Also

$$H_j(\mathbb{RP}^2) = 0 \quad (j \geq 3).$$

Generators for the projective plane.

- $H_0(\mathbb{RP}^2) \cong \mathbb{Z}$ is generated by the class of any vertex.
- $H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$ is generated by the class $[a]$ of the basic loop in the quotient-square model. It satisfies

$$2[a] = 0.$$

- Since $\mathbb{RP}^2$ is nonorientable,

$$H_2(\mathbb{RP}^2) = 0.$$

So

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2, \quad H_2(\mathbb{RP}^2) = 0.$$

1.4 Final answer

We conclude that

$H_0(T) \cong \mathbb{Z},$	$H_1(T) \cong \mathbb{Z}^2,$	$H_2(T) \cong \mathbb{Z},$
$H_0(K) \cong \mathbb{Z},$	$H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2,$	$H_2(K) = 0,$
$H_0(\mathbb{RP}^2) \cong \mathbb{Z},$	$H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2,$	$H_2(\mathbb{RP}^2) = 0,$

and all higher homology groups vanish.

Generators may be chosen as follows:

- H_0 : the class of any vertex;
- $H_1(T)$: the two basic loops a and b ;
- $H_2(T)$: the fundamental class, represented by the sum of all oriented 2-simplices;
- $H_1(K)$: one untwisted loop of infinite order and one twisted loop of order 2;
- $H_1(\mathbb{RP}^2)$: the basic projective loop of order 2.

2 Problem 7 Top homology of an n -dimensional pseudomanifold type simplicial complex

2.1 Definitions and background

Top-dimensional simplices and adjacency. Let K be an n -dimensional simplicial complex. The top homology group $H_n(K)$ is determined entirely by the n -cycles, because there are no $(n + 1)$ -simplices and hence

$$B_n(K) = 0.$$

Thus

$$H_n(K) = Z_n(K) = \ker(\partial_n : C_n(K) \rightarrow C_{n-1}(K)).$$

Geometric intuition. The hypotheses in this problem say that:

- every $(n - 1)$ -simplex lies on exactly two n -simplices;
- the n -simplices form one connected family under adjacency along $(n - 1)$ -faces.

This is the simplicial analogue of a connected closed n -dimensional pseudomanifold. One expects the top homology to be either $\mathbb{Z}$ (the orientable case) or 0 (the nonorientable case).

2.2 Problem statement

Problem. Let K be an n -dimensional simplicial complex such that

- (i) every $(n - 1)$ -simplex is a face of precisely two n -simplices, and
- (ii) every pair of n -simplices can be connected by a sequence of n -simplices such that adjacent terms share an $(n - 1)$ -dimensional face.

Show that

$$H_n(K)$$

is either $\mathbb{Z}$ or trivial. In the first case show that $H_n(K)$ is generated by a cycle which is a sum of all n -simplices with suitable orientations.

2.3 Solution

Let the n -simplices of K be

$$\sigma_1, \dots, \sigma_r.$$

Then

$$C_n(K) \cong \bigoplus_{i=1}^r \mathbb{Z} \sigma_i.$$

Since K has dimension n , there are no $(n + 1)$ -simplices. Therefore

$$B_n(K) = 0,$$

so

$$H_n(K) = Z_n(K) = \ker(\partial_n).$$

Thus it is enough to determine all n -cycles.

Step 1: a general n -cycle.

Take an arbitrary n -chain

$$z = \sum_{i=1}^r a_i \sigma_i \quad (a_i \in \mathbb{Z}).$$

Assume that z is a cycle:

$$\partial z = 0.$$

Let τ be any $(n - 1)$ -simplex of K . By hypothesis (i), τ is a face of exactly two n -simplices, say σ_i and σ_j . In the boundary ∂z , the coefficient of τ is

$$a_i[\tau : \sigma_i] + a_j[\tau : \sigma_j],$$

where each incidence number is ± 1 . Since $\partial z = 0$, this coefficient must vanish:

$$a_i[\tau : \sigma_i] + a_j[\tau : \sigma_j] = 0.$$

Hence

$$a_j = \pm a_i.$$

So whenever two n -simplices share an $(n - 1)$ -face, the absolute values of their coefficients in a cycle are equal.

Step 2: connectedness forces all absolute values to agree.

By hypothesis (ii), any two n -simplices can be connected by a chain of adjacent n -simplices

$$\sigma_{i_0}, \sigma_{i_1}, \dots, \sigma_{i_m}$$

such that consecutive simplices share an $(n - 1)$ -face. Repeatedly applying the conclusion of Step 1 along this chain, we obtain

$$|a_{i_0}| = |a_{i_1}| = \dots = |a_{i_m}|.$$

Since any pair of n -simplices can be connected in this way, it follows that all coefficients of the cycle satisfy

$$|a_1| = |a_2| = \dots = |a_r|.$$

Thus every n -cycle is of the form

$$z = m \sum_{i=1}^r \varepsilon_i \sigma_i,$$

where

$$m \in \mathbb{Z}, \quad \varepsilon_i \in \{\pm 1\}.$$

Therefore the group of n -cycles has rank at most 1. Since

$$H_n(K) = Z_n(K),$$

it follows that $H_n(K)$ is either trivial or isomorphic to $\mathbb{Z}$.

This proves the first assertion.

Step 3: if $H_n(K) \neq 0$, construct a generator.

Assume now that $H_n(K) \neq 0$. Then there exists a nonzero n -cycle

$$z = \sum_{i=1}^r a_i \sigma_i.$$

By the previous step, all $|a_i|$ are equal. Since $z \neq 0$, let

$$|a_i| = m > 0 \quad \text{for all } i.$$

Then

$$z = m \sum_{i=1}^r \varepsilon_i \sigma_i \quad (\varepsilon_i = \pm 1).$$

Now orient each simplex σ_i so that its sign agrees with ε_i . With these chosen orientations, the cycle takes the form

$$z = m(\sigma_1 + \cdots + \sigma_r).$$

Therefore the chain

$$c = \sigma_1 + \cdots + \sigma_r$$

(with suitable orientations) is itself an n -cycle.

Since every n -cycle is an integer multiple of this one, the cycle c generates $Z_n(K)$, and hence generates $H_n(K) \cong \mathbb{Z}$.

Thus, in the nontrivial case, $H_n(K)$ is generated by the sum of all n -simplices with suitably chosen orientations.

Step 4: geometric interpretation.

The two possibilities correspond to orientability behavior.

- If one can choose orientations on the n -simplices so that along each shared $(n-1)$ -face the contributions cancel, then the sum of all n -simplices is a nonzero cycle. In that case

$$H_n(K) \cong \mathbb{Z}.$$

- If no such global consistent choice exists, then there is no nonzero n -cycle at all, and

$$H_n(K) = 0.$$

So this theorem is the simplicial version of the familiar statement that the top homology of a connected closed n -dimensional pseudomanifold is either $\mathbb{Z}$ or 0, depending on orientability.

2.4 Conclusion

We have proved that

$$H_n(K) \text{ is either } 0 \text{ or } \mathbb{Z}.$$

Moreover, if

$$H_n(K) \cong \mathbb{Z},$$

then a generator is represented by

$$\sum_{\sigma \in K^{(n)}} \sigma,$$

where the sum runs over all n -simplices and the orientations are chosen so that adjacent simplices cancel along their common $(n-1)$ -faces.

Thus the top homology is completely controlled by whether the top-dimensional simplices admit a globally compatible orientation.

2.5 Theory, definitions, and examples for the homology of triangulated surfaces and top-dimensional cycles

The previous two problems illustrate two closely related themes:

- how to compute homology groups from a triangulation, and
- how top-dimensional homology detects orientability.

We collect here the main definitions, ideas, and examples behind those computations.

Simplicial chains. If K is a simplicial complex, the group of n -chains is the free abelian group

$$C_n(K)$$

generated by the oriented n -simplices of K . Thus an n -chain is a finite formal sum

$$c = \sum_i a_i \sigma_i, \quad a_i \in \mathbb{Z},$$

where each σ_i is an oriented n -simplex.

Boundary map. There is a homomorphism

$$\partial_n : C_n(K) \rightarrow C_{n-1}(K)$$

defined on an oriented simplex by the alternating sum of its codimension-one faces. For example,

$$\partial[v_0 v_1] = [v_1] - [v_0],$$

and

$$\partial[v_0 v_1 v_2] = [v_1 v_2] - [v_0 v_2] + [v_0 v_1].$$

The fundamental identity

$$\partial_{n-1} \partial_n = 0$$

says that the boundary of a boundary is always zero.

Cycles, boundaries, and homology. An n -chain $z \in C_n(K)$ is an n -cycle if

$$\partial z = 0.$$

An n -chain is an n -boundary if

$$z = \partial c$$

for some $(n + 1)$ -chain c . We write

$$Z_n(K) = \ker \partial_n, \quad B_n(K) = \operatorname{im} \partial_{n+1}.$$

Since $\partial^2 = 0$, every boundary is a cycle:

$$B_n(K) \subseteq Z_n(K).$$

The n -th homology group is

$$H_n(K) = Z_n(K)/B_n(K).$$

Thus homology measures cycles modulo those cycles which already bound something.

Interpretation of low-dimensional homology. The first few homology groups have standard geometric meanings:

- $H_0(K)$ measures connected components;
- $H_1(K)$ measures 1-dimensional holes, or independent loop classes;
- $H_2(K)$ measures 2-dimensional voids or shells;
- more generally, $H_n(K)$ measures n -dimensional cycles that do not bound in dimension $n + 1$.

Generators. If $H_n(K) \cong \mathbb{Z}$, a generator is a homology class corresponding to $1 \in \mathbb{Z}$. If $H_n(K) \cong \mathbb{Z}^2$, one must specify two independent generators. If

$$H_n(K) \cong \mathbb{Z}/m,$$

then a generator is a class of order m .

In the triangulated surface examples, generators can often be represented by very concrete cycles built from edges or triangles.

Torsion. An element $x \in H_n(K)$ is called a torsion element if

$$mx = 0$$

for some nonzero integer m . For example, if

$$H_1(K) \cong \mathbb{Z}/2,$$

then the nonzero class is a torsion element of order 2.

This is what happens in the projective plane and in one part of the Klein bottle homology.

The torus. The torus T is connected and orientable. Its homology groups are

$$H_0(T) \cong \mathbb{Z}, \quad H_1(T) \cong \mathbb{Z} \oplus \mathbb{Z}, \quad H_2(T) \cong \mathbb{Z}.$$

Geometrically:

- $H_0(T)$ is generated by the class of any vertex;
- $H_1(T)$ is generated by the two fundamental loops, one going around the horizontal direction and one around the vertical direction;
- $H_2(T)$ is generated by the fundamental class of the whole oriented surface.

In a triangulation, that fundamental 2-cycle is represented by the sum of all 2-simplices with compatible orientations.

The Klein bottle. The Klein bottle K is connected but nonorientable. Its homology groups are

$$H_0(K) \cong \mathbb{Z}, \quad H_1(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2, \quad H_2(K) = 0.$$

Thus:

- $H_0(K)$ is generated by the class of any vertex;
- $H_1(K)$ has one infinite-order generator, represented by the untwisted loop direction;
- $H_1(K)$ also has one element of order 2, represented by the twisted direction;
- there is no nonzero 2-dimensional fundamental class, because the Klein bottle is nonorientable.

The projective plane. The projective plane $\mathbb{RP}^2$ is connected and nonorientable. Its homology groups are

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2, \quad H_2(\mathbb{RP}^2) = 0.$$

Here:

- H_0 is generated by the class of any vertex;
- H_1 is generated by the basic projective loop, which has order 2;
- H_2 vanishes because $\mathbb{RP}^2$ is nonorientable.

Orientability and top homology. The most important conceptual point behind Problem 7 is that top-dimensional homology detects orientability.

If M is a connected closed orientable n -manifold, then

$$H_n(M) \cong \mathbb{Z}.$$

A generator is the *fundamental class*

$$[M],$$

represented by the whole manifold with a global orientation.

If M is connected, closed, and nonorientable, then

$$H_n(M) = 0.$$

The reason is that one cannot choose orientations on the top-dimensional simplices consistently throughout the triangulation so that codimension-one faces cancel everywhere.

This is exactly why:

$$H_2(T) \cong \mathbb{Z}, \quad H_2(K) = 0, \quad H_2(\mathbb{RP}^2) = 0.$$

Why the top homology in Problem 7 is only 0 or $\mathbb{Z}$. Suppose K is n -dimensional and satisfies:

1. every $(n - 1)$ -simplex is a face of exactly two n -simplices;
2. every pair of n -simplices is connected by a chain of adjacent n -simplices sharing $(n - 1)$ -faces.

Then K behaves like a connected closed n -dimensional pseudomanifold.

Take an n -cycle

$$z = \sum_i a_i \sigma_i.$$

Each $(n - 1)$ -face lies in exactly two n -simplices, so cancellation of that face in ∂z forces the coefficients of those two simplices to agree up to sign. Because all n -simplices are connected by adjacency, this propagates through the entire complex and implies that every coefficient a_i has the same absolute value.

Therefore every n -cycle is an integer multiple of a single global pattern:

$$z = m \sum_i \varepsilon_i \sigma_i, \quad \varepsilon_i \in \{\pm 1\}.$$

So the group of n -cycles has rank at most 1. Since there are no $(n + 1)$ -simplices,

$$B_n(K) = 0,$$

and thus

$$H_n(K) = Z_n(K).$$

Hence

$$H_n(K) \text{ is either } 0 \text{ or } \mathbb{Z}.$$

Generator in the nontrivial case. If $H_n(K) \neq 0$, then there exists a nonzero n -cycle. Reorienting the n -simplices suitably, one obtains a cycle of the form

$$\sigma_1 + \sigma_2 + \cdots + \sigma_r,$$

where the sum is taken over all n -simplices of K with compatible orientations. This cycle generates $H_n(K) \cong \mathbb{Z}$.

So in the nontrivial case, the generator of top homology is the simplicial version of the fundamental class.

Examples illustrating Problem 7.

- *Boundary of a tetrahedron.* This triangulates S^2 . Every edge belongs to exactly two triangles, and all triangles are connected by adjacency. Since S^2 is orientable,

$$H_2(S^2) \cong \mathbb{Z}.$$

A generator is the sum of the four triangular faces with suitable orientations.

- *Triangulated torus.* Again every edge belongs to exactly two triangles, and all triangles are connected. Since the torus is orientable,

$$H_2(T) \cong \mathbb{Z}.$$

A generator is the sum of all oriented 2-simplices.

- *Triangulated Klein bottle.* The same adjacency hypotheses hold, but the Klein bottle is nonorientable, so

$$H_2(K) = 0.$$

- *Triangulated projective plane.* Likewise, the hypotheses hold, but $\mathbb{RP}^2$ is nonorientable, so

$$H_2(\mathbb{RP}^2) = 0.$$

- *Boundary of an $(n + 1)$ -simplex.* This triangulates S^n . Every $(n - 1)$ -face lies in exactly two n -faces, and all n -faces are adjacent through codimension-one faces. Since S^n is orientable,

$$H_n(S^n) \cong \mathbb{Z}.$$

A generator is the sum of all n -faces with compatible orientations.

Summary table for closed triangulated surfaces. For the main surface examples one has:

Surface	Orientable?	Top homology
S^2	yes	$H_2 \cong \mathbb{Z}$
T^2	yes	$H_2 \cong \mathbb{Z}$
K	no	$H_2 = 0$
$\mathbb{RP}^2$	no	$H_2 = 0$

Main principle. The general rule is:

$$\text{top homology} \longleftrightarrow \text{orientability}.$$

If the top-dimensional simplices admit a global compatible orientation, then the sum of all of them is a nontrivial cycle generating a copy of $\mathbb{Z}$. If they do not, then the top homology vanishes.

Thus Problems 6 and 7 together show both how to compute homology of specific triangulated surfaces and why the answer is governed, in top degree, by orientability.

3 Problem 8 Acyclic chain complexes of free abelian groups are chain contractible

3.1 Definitions and background

Chain homotopy. Let

$$(C_\bullet, d) \quad \text{and} \quad (D_\bullet, \partial)$$

be chain complexes, and let

$$f, g : C_\bullet \rightarrow D_\bullet$$

be chain maps. A *chain homotopy* from f to g is a family of homomorphisms

$$s_n : C_n \rightarrow D_{n+1}$$

such that

$$f_n - g_n = \partial_{n+1}s_n + s_{n-1}d_n \quad \text{for all } n.$$

In particular, saying that

$$\text{id}_{C_\bullet} \simeq 0$$

means that there exist homomorphisms

$$s_n : C_n \rightarrow C_{n+1}$$

such that

$$d_{n+1}s_n + s_{n-1}d_n = \text{id}_{C_n} \quad \text{for all } n.$$

Such a family (s_n) is called a *contracting homotopy*.

Acyclicity. A chain complex $(C_\bullet, d)$ is called *acyclic* if

$$H_n(C_\bullet) = 0 \quad \text{for all } n.$$

Equivalently,

$$Z_n(C_\bullet) = B_n(C_\bullet) \quad \text{for all } n,$$

that is, every cycle is a boundary.

Splitting over free abelian groups. A basic fact about finitely-generated free abelian groups is that every subgroup is again free abelian. In particular, if

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

is a short exact sequence of abelian groups and C is free abelian, then the sequence splits. This will be the key algebraic input in the proof.

3.2 Problem statement

Problem. Suppose that $(C_\bullet, d)$ is a chain complex of finitely-generated free abelian groups with

$$H_*(C_\bullet, d) = 0.$$

Show that

$$\text{id} : C_\bullet \rightarrow C_\bullet$$

is chain homotopic to the zero map.

3.3 Solution

We must show that there exist homomorphisms

$$s_n : C_n \rightarrow C_{n+1}$$

such that

$$d_{n+1}s_n + s_{n-1}d_n = \text{id}_{C_n} \quad \text{for all } n.$$

Step 1: use acyclicity.

Since the homology vanishes, for every n we have

$$H_n(C_\bullet) = 0.$$

Thus

$$Z_n = \ker d_n = \text{im } d_{n+1} = B_n.$$

So every cycle is a boundary.

Now consider the sequence

$$0 \longrightarrow B_n \longrightarrow C_n \xrightarrow{d_n} B_{n-1} \longrightarrow 0.$$

This sequence is exact, because:

- $\ker d_n = Z_n = B_n$,
- $\text{im } d_n = B_{n-1}$.

Since C_n is finitely-generated free abelian, the subgroup $B_n \subseteq C_n$ is also free abelian. Hence B_{n-1} is free abelian as well, and the exact sequence splits. Therefore for each n there exists a subgroup $D_n \subseteq C_n$ such that

$$C_n = B_n \oplus D_n,$$

and the restriction

$$d_n|_{D_n} : D_n \rightarrow B_{n-1}$$

is an isomorphism.

Step 2: define the contracting homotopy.

We now define

$$s_{n-1} : C_{n-1} \rightarrow C_n$$

for each n .

Since

$$C_{n-1} = B_{n-1} \oplus D_{n-1},$$

every element $x \in C_{n-1}$ can be written uniquely as

$$x = b + d, \quad b \in B_{n-1}, \quad d \in D_{n-1}.$$

Define

$$s_{n-1}(x) = (d_n|_{D_n})^{-1}(b).$$

In other words:

- on the boundary part B_{n-1} , the map s_{n-1} is the inverse of $d_n|_{D_n}$;
- on the complementary part D_{n-1} , the map s_{n-1} is zero.

So explicitly,

$$s_{n-1}|_{B_{n-1}} = (d_n|_{D_n})^{-1}, \quad s_{n-1}|_{D_{n-1}} = 0.$$

Step 3: verify the homotopy identity.

Take $x \in C_n$. Write

$$x = b + c, \quad b \in B_n, \quad c \in D_n.$$

We compute

$$(d_{n+1}s_n + s_{n-1}d_n)(x).$$

First,

$$s_n(x) = s_n(b + c) = s_n(b),$$

because s_n vanishes on D_n . Since $b \in B_n$, the element $s_n(b) \in D_{n+1}$ is chosen so that

$$d_{n+1}s_n(b) = b.$$

Hence

$$d_{n+1}s_n(x) = b.$$

Next,

$$d_n(x) = d_n(b + c) = d_n(c),$$

because $b \in B_n = \ker d_n$. Since $c \in D_n$, we have

$$d_n(c) \in B_{n-1},$$

and by definition of s_{n-1} ,

$$s_{n-1}d_n(x) = s_{n-1}(d_n(c)) = c.$$

Therefore

$$(d_{n+1}s_n + s_{n-1}d_n)(x) = b + c = x.$$

Since this holds for every $x \in C_n$, we get

$$d_{n+1}s_n + s_{n-1}d_n = \text{id}_{C_n}.$$

Thus

$$\text{id}_{C_\bullet} \simeq 0.$$

3.4 Conclusion

We have constructed homomorphisms

$$s_n : C_n \rightarrow C_{n+1}$$

such that

$$d_{n+1}s_n + s_{n-1}d_n = \text{id}_{C_n} \quad \text{for all } n.$$

Hence the identity chain map is chain homotopic to the zero map.

So every acyclic chain complex of finitely-generated free abelian groups is *chain contractible*.

4 Problem 9 Products of polyhedra are polyhedra of simplicial complexes

4.1 Definitions and background

Polyhedron of a simplicial complex. If $K \subseteq \mathbb{R}^m$ is a simplicial complex, its polyhedron is

$$|K| = \bigcup_{\sigma \in K} \sigma.$$

Goal. Given simplicial complexes

$$K \subseteq \mathbb{R}^m, \quad L \subseteq \mathbb{R}^n,$$

we want to show that the product

$$|K| \times |L| \subseteq \mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$$

is again the polyhedron of some simplicial complex.

Key construction. The essential step is to triangulate the product of two simplices. The standard triangulation is called the *staircase triangulation*. Once this is done, the general case follows by applying the construction simplex-by-simplex and checking that the triangulations agree on overlaps.

4.2 Problem statement

Problem. For simplicial complexes K and L inside $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively, show that

$$|K| \times |L| \subseteq \mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$$

is the polyhedron of a simplicial complex.

Hint: Prove it first in the case where both K and L consist of a single simplex (plus all its faces).

4.3 Solution

We first treat the case of two simplices.

Step 1: product of two simplices.

Suppose

$$K = \{\text{all faces of a simplex } \sigma\}, \quad L = \{\text{all faces of a simplex } \tau\},$$

where

$$\sigma = \langle v_0, \dots, v_p \rangle \subseteq \mathbb{R}^m, \quad \tau = \langle w_0, \dots, w_q \rangle \subseteq \mathbb{R}^n.$$

Then

$$|\sigma| \times |\tau| \subseteq \mathbb{R}^{m+n}$$

has vertices

$$(v_i, w_j) \in \mathbb{R}^m \times \mathbb{R}^n \quad (0 \leq i \leq p, 0 \leq j \leq q).$$

We triangulate $\sigma \times \tau$ using these vertices.

Step 2: monotone lattice paths and staircase simplices.

Consider the rectangular grid of index pairs

$$(i, j), \quad 0 \leq i \leq p, 0 \leq j \leq q.$$

A *monotone lattice path* from $(0, 0)$ to (p, q) is a path that moves only:

- one step to the right:

$$(i, j) \rightarrow (i + 1, j),$$

- or one step upward:

$$(i, j) \rightarrow (i, j + 1).$$

Such a path has exactly $p + q + 1$ vertices:

$$(i_0, j_0), (i_1, j_1), \dots, (i_{p+q}, j_{p+q}).$$

To such a path P , associate the simplex

$$\Delta_P = \langle (v_{i_0}, w_{j_0}), (v_{i_1}, w_{j_1}), \dots, (v_{i_{p+q}}, w_{j_{p+q}}) \rangle \subseteq \mathbb{R}^{m+n}.$$

Each Δ_P is a simplex of dimension $p + q$.

The collection of all such simplices Δ_P , as P ranges over all monotone lattice paths from $(0, 0)$ to (p, q) , is called the *staircase triangulation* of $\sigma \times \tau$.

Step 3: why this triangulates the product.

The staircase simplices cover the whole product $\sigma \times \tau$, and any two of them intersect in a common face. This is a standard fact about the staircase construction.

Geometrically, one may think of $\sigma \times \tau$ as subdivided according to the relative order of the barycentric coordinates in the two factors. The monotone lattice paths record these possible orderings, and each such ordering determines one simplex.

Thus $\sigma \times \tau$ is the polyhedron of a simplicial complex.

Example 1: $\Delta^1 \times \Delta^1$.

The product of two edges is a square. The staircase triangulation divides it into two triangles, corresponding to the two monotone lattice paths from $(0, 0)$ to $(1, 1)$:

$$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1), \quad (0, 0) \rightarrow (0, 1) \rightarrow (1, 1).$$

So the square is triangulated by a diagonal.

Example 2: $\Delta^1 \times \Delta^2$.

The product of an edge and a triangle is a triangular prism. The staircase triangulation subdivides it into 3 tetrahedra, corresponding to the 3 monotone paths from $(0,0)$ to $(1,2)$.

Step 4: pass to general simplicial complexes.

Now let $K \subseteq \mathbb{R}^m$ and $L \subseteq \mathbb{R}^n$ be arbitrary simplicial complexes. Then

$$|K| \times |L| = \bigcup_{\sigma \in K, \tau \in L} (\sigma \times \tau).$$

For each pair (σ, τ) , triangulate $\sigma \times \tau$ by the staircase construction above.

We claim that these triangulations fit together to form a simplicial complex on the whole product.

Indeed, if

$$\sigma' \leq \sigma, \quad \tau' \leq \tau,$$

then

$$\sigma' \times \tau' \subseteq \sigma \times \tau,$$

and the staircase triangulation of $\sigma' \times \tau'$ is exactly the restriction of the staircase triangulation of $\sigma \times \tau$. Therefore the triangulations agree on common faces.

Thus the simplices coming from all products $\sigma \times \tau$ together form a simplicial complex $M \subseteq \mathbb{R}^{m+n}$.

Its polyhedron is exactly

$$|M| = |K| \times |L|.$$

Example 3: $S^1 \times I$.

If K triangulates a circle and L triangulates an interval, then

$$|K| \times |L| \cong S^1 \times I$$

is a cylinder. The theorem shows that this cylinder is the polyhedron of a simplicial complex.

Example 4: $S^1 \times S^1$.

If K and L both triangulate S^1 , then

$$|K| \times |L| \cong S^1 \times S^1 = T^2.$$

So the torus admits a triangulation obtained from a product triangulation.

4.4 Conclusion

We have shown that the product of two simplices admits a staircase triangulation, and that these triangulations are compatible on faces. Therefore for arbitrary simplicial complexes $K \subseteq \mathbb{R}^m$ and $L \subseteq \mathbb{R}^n$, the product

$$|K| \times |L| \subseteq \mathbb{R}^{m+n}$$

is the polyhedron of a simplicial complex.

Hence

$$|K| \times |L| \text{ is the polyhedron of a simplicial complex.}$$

4.5 Theory, definitions, and examples for acyclic chain complexes and products of polyhedra

We now collect the main definitions, ideas, and examples behind Problems 8 and 9.

Chain complexes. A *chain complex* of abelian groups is a sequence

$$\cdots \xrightarrow{d_{n+2}} C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} \cdots$$

such that

$$d_n \circ d_{n+1} = 0 \quad \text{for all } n.$$

The condition $d^2 = 0$ means that the boundary of a boundary is always zero.

Cycles, boundaries, and homology. For a chain complex $(C_\bullet, d)$, define

$$Z_n = \ker d_n \quad (\text{the } n\text{-cycles}),$$

and

$$B_n = \operatorname{im} d_{n+1} \quad (\text{the } n\text{-boundaries}).$$

Because $d^2 = 0$, one always has

$$B_n \subseteq Z_n.$$

The n -th homology group is

$$H_n(C_\bullet) = Z_n / B_n.$$

Thus homology measures cycles modulo those cycles which already bound something.

Acyclic complexes. A chain complex is called *acyclic* if all of its homology groups vanish:

$$H_n(C_\bullet) = 0 \quad \text{for all } n.$$

Equivalently,

$$Z_n = B_n \quad \text{for all } n.$$

So an acyclic complex is one in which every cycle is already a boundary.

Chain homotopy. If

$$f, g : C_\bullet \rightarrow D_\bullet$$

are chain maps, a *chain homotopy* from f to g is a family of homomorphisms

$$s_n : C_n \rightarrow D_{n+1}$$

such that

$$f_n - g_n = \partial_{n+1}s_n + s_{n-1}d_n \quad \text{for all } n.$$

In particular, saying that

$$\mathrm{id}_{C_\bullet} \simeq 0$$

means that there exist maps

$$s_n : C_n \rightarrow C_{n+1}$$

such that

$$d_{n+1}s_n + s_{n-1}d_n = \mathrm{id}_{C_n}.$$

Such a family (s_n) is called a *contracting homotopy*, and the complex is called *chain contractible*.

The meaning of Problem 8. Problem 8 says that if $(C_\bullet, d)$ is a chain complex of finitely-generated free abelian groups and

$$H_*(C_\bullet) = 0,$$

then the complex is not only acyclic, but actually chain contractible:

$$\mathrm{id}_{C_\bullet} \simeq 0.$$

This is stronger than the vanishing of homology. Homology only says that every cycle bounds; chain contractibility says that one can build an explicit algebraic contraction of the entire complex.

Why freeness matters. The proof of Problem 8 uses the fact that every subgroup of a finitely-generated free abelian group is again free abelian. Therefore if

$$0 \rightarrow B_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$$

is exact, then the sequence splits, so one may write

$$C_n \cong B_n \oplus D_n$$

for some complement D_n , with

$$d_n|_{D_n} : D_n \rightarrow B_{n-1}$$

an isomorphism.

This splitting allows one to define a contracting homotopy explicitly. So the underlying algebraic principle is:

$$\text{acyclic} + \text{free chain groups} \implies \text{chain contractible}.$$

Example 1: the simplest acyclic complex. Consider the chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0,$$

with the nonzero groups in degrees 1 and 0. Since the boundary map is the identity, we have

$$H_1 = 0, \quad H_0 = 0.$$

So the complex is acyclic.

A contracting homotopy is immediate: define

$$s_0 : \mathbb{Z} \rightarrow \mathbb{Z}$$

to be the identity, and all other $s_n = 0$. Then

$$d_1 s_0 = \text{id}_{C_0}, \quad s_0 d_1 = \text{id}_{C_1}.$$

So indeed

$$ds + sd = \text{id}.$$

Example 2: a longer acyclic complex. Consider

$$0 \rightarrow \mathbb{Z} \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} \mathbb{Z}^2 \xrightarrow{(0 \ 1)} \mathbb{Z} \rightarrow 0.$$

Placed in degrees 2, 1, 0, this is a chain complex because

$$(0 \ 1) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0.$$

Its homology vanishes:

- $H_2 = 0$ because $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is injective;
- $H_1 = 0$ because $\ker(0 \ 1) = \{(a, 0)\} = \text{im} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$;
- $H_0 = 0$ because $(0 \ 1)$ is surjective.

Since all groups are free abelian, Problem 8 applies and shows that this chain complex is chain contractible.

Example 3: simplicial chains of a simplex. The augmented simplicial chain complex of a simplex is acyclic. In fact, it admits an explicit chain contraction. This is one of the classical geometric sources of Problem 8: algebraically, the simplex is contractible, so its simplicial chain complex is chain contractible.

Polyhedra and simplicial complexes. If $K \subseteq \mathbb{R}^m$ is a simplicial complex, its *polyhedron* is

$$|K| = \bigcup_{\sigma \in K} \sigma.$$

Problem 9 asks whether the product of two such polyhedra is again the polyhedron of some simplicial complex.

Why this is not automatic. Even if σ and τ are simplices, their product

$$\sigma \times \tau$$

is usually not itself a simplex. For example:

- $\Delta^1 \times \Delta^1$ is a square;
- $\Delta^1 \times \Delta^2$ is a triangular prism;
- $\Delta^2 \times \Delta^2$ is a 4-dimensional polytope.

So one cannot simply take products of simplices and expect to obtain a simplicial complex. One must subdivide each product into simplices.

Staircase triangulation. The standard way to triangulate a product of simplices is the *staircase triangulation*. If

$$\sigma = \langle v_0, \dots, v_p \rangle, \quad \tau = \langle w_0, \dots, w_q \rangle,$$

then the vertices of $\sigma \times \tau$ are the pairs

$$(v_i, w_j).$$

The simplices in the triangulation correspond to monotone lattice paths from $(0, 0)$ to (p, q) in the grid

$$\{0, \dots, p\} \times \{0, \dots, q\},$$

where each step moves either one unit to the right or one unit upward.

Each such path determines a simplex whose vertices are the corresponding points (v_i, w_j) . The collection of all such simplices triangulates $\sigma \times \tau$.

Example 4: $\Delta^1 \times \Delta^1$. The product of two intervals is a square:

$$\Delta^1 \times \Delta^1 = [0, 1] \times [0, 1].$$

This is not a simplex, but it can be triangulated by a diagonal into two triangles. These two triangles correspond to the two monotone lattice paths from $(0, 0)$ to $(1, 1)$:

$$(0, 0) \rightarrow (1, 0) \rightarrow (1, 1), \quad (0, 0) \rightarrow (0, 1) \rightarrow (1, 1).$$

Example 5: $\Delta^1 \times \Delta^2$. The product of an interval and a triangle is a triangular prism. This can be triangulated into 3 tetrahedra, corresponding to the 3 monotone paths from $(0, 0)$ to $(1, 2)$.

So the prism is not a simplex, but it is the polyhedron of a simplicial complex after subdivision.

From simplices to general complexes. If $K \subseteq \mathbb{R}^m$ and $L \subseteq \mathbb{R}^n$ are arbitrary simplicial complexes, then

$$|K| \times |L| = \bigcup_{\sigma \in K, \tau \in L} (\sigma \times \tau).$$

Each product $\sigma \times \tau$ can be triangulated by the staircase construction. The key compatibility fact is that if

$$\sigma' \leq \sigma, \quad \tau' \leq \tau,$$

then the triangulation of

$$\sigma' \times \tau'$$

is exactly the restriction of the triangulation of

$$\sigma \times \tau.$$

Therefore the local triangulations agree on overlaps and glue together to form one global simplicial complex whose polyhedron is

$$|K| \times |L|.$$

Example 6: the cylinder. If K triangulates S^1 and L triangulates the interval $I = [0, 1]$, then

$$|K| \times |L| \cong S^1 \times I$$

is a cylinder. Problem 9 says that this cylinder is the polyhedron of a simplicial complex.

Example 7: the torus. If K and L both triangulate S^1 , then

$$|K| \times |L| \cong S^1 \times S^1 = T^2.$$

Thus the torus admits a triangulation coming from a product triangulation. This is one of the most important examples of the theorem.

Example 8: cubes and higher products. The square

$$I \times I$$

and the cube

$$I \times I \times I$$

can both be triangulated. The square is subdivided by one diagonal, and the cube may be subdivided into simplices in the analogous higher-dimensional way. This is a higher-dimensional version of the product triangulation idea.

Relationship with CW complexes. In general topology, products of CW complexes are again CW complexes under suitable hypotheses. Problem 9 is the simplicial analogue of that principle: the category of polyhedra is closed under products.

So the theorem says that one may remain inside the simplicial world when forming products of triangulated spaces.

Main conceptual summary. Problems 8 and 9 illustrate two complementary themes.

- Problem 8 is an algebraic statement:

$$\text{acyclic free chain complex} \implies \text{chain contractible}.$$

It says that in the free setting, vanishing homology is strong enough to build an explicit chain-level contraction.

- Problem 9 is a geometric statement:

$$\text{product of polyhedra} \implies \text{again a polyhedron}.$$

It says that the simplicial category is closed under products after suitable subdivision.

Thus one problem gives strong algebraic control of chain complexes, while the other gives strong geometric control of products of triangulated spaces.

5 Problem 1 Exact sequences of abelian groups: determining A and α

Statement

For each of the following exact sequences of abelian groups and homomorphisms, determine as much as possible about the unknown group A and the homomorphism α :

- (i) $0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z} \longrightarrow 0,$
- (ii) $0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0,$
- (iii) $0 \longrightarrow A \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0,$
- (iv) $0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow 0.$

Solution.

We analyze the four cases separately.

(i) Consider

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z} \longrightarrow 0.$$

This is a short exact sequence

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z} \longrightarrow 0.$$

Since $\mathbb{Z}$ is a free abelian group, every short exact sequence ending in $\mathbb{Z}$ splits. Hence

$$A \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

So in this case

$$\boxed{A \cong \mathbb{Z} \oplus \mathbb{Z}/2.}$$

(ii) Now consider

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Here A is a finite abelian group of order 4. Up to isomorphism, there are exactly two abelian groups of order 4:

$$\mathbb{Z}/4 \quad \text{and} \quad \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Both can occur:

- the split extension gives $A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2,$
- a non-split extension gives $A \cong \mathbb{Z}/4.$

Therefore the exact sequence does not determine A uniquely, but it does determine that

$$\boxed{A \cong \mathbb{Z}/4 \quad \text{or} \quad A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2.}$$

(iii) Consider

$$0 \longrightarrow A \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Any homomorphism $\alpha : \mathbb{Z} \rightarrow \mathbb{Z}$ is multiplication by some integer n :

$$\alpha(k) = nk.$$

Its cokernel is

$$\text{coker}(\alpha) \cong \mathbb{Z}/n\mathbb{Z}.$$

By exactness, this cokernel must be isomorphic to $\mathbb{Z}/2$, so necessarily

$$|n| = 2.$$

Thus

$$\alpha = \times 2 \quad \text{or} \quad \alpha = \times (-2).$$

Now the kernel of multiplication by ± 2 on $\mathbb{Z}$ is zero, hence

$$A = \ker(\alpha) = 0.$$

Therefore

$$\boxed{A = 0, \quad \alpha = \times 2 \text{ or } \times (-2).}$$

(iv) Finally consider

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow 0.$$

First note that any homomorphism

$$\mathbb{Z}/2 \longrightarrow \mathbb{Z}$$

must be the zero homomorphism, because $\mathbb{Z}$ has no nonzero element of order 2. Therefore the map

$$\mathbb{Z}/2 \longrightarrow \mathbb{Z}$$

is zero.

Exactness at the copy of $\mathbb{Z}$ before α now gives

$$\ker(\alpha) = \text{im}(\mathbb{Z}/2 \rightarrow \mathbb{Z}) = 0,$$

so α is injective. Exactness at the last $\mathbb{Z}$ says α is surjective. Hence α is an automorphism of $\mathbb{Z}$, so

$$\alpha = \pm \text{id}_{\mathbb{Z}}.$$

Since the map $\mathbb{Z}/2 \rightarrow \mathbb{Z}$ is zero, exactness also gives that

$$A \longrightarrow \mathbb{Z}/2$$

is surjective, and its kernel is the image of $\mathbb{Z}/3$. Thus we get a short exact sequence

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

So A is an abelian group of order 6. Every abelian group of order 6 is cyclic, hence

$$A \cong \mathbb{Z}/6.$$

Therefore

$$A \cong \mathbb{Z}/6, \quad \alpha = \pm \text{id}_{\mathbb{Z}}.$$

Collecting the answers:

$$(i) \ A \cong \mathbb{Z} \oplus \mathbb{Z}/2,$$

$$(ii) \ A \cong \mathbb{Z}/4 \text{ or } \mathbb{Z}/2 \oplus \mathbb{Z}/2,$$

$$(iii) \ A = 0, \ \alpha = \times 2 \text{ or } \times (-2),$$

$$(iv) \ A \cong \mathbb{Z}/6, \ \alpha = \pm \text{id}_{\mathbb{Z}}.$$

5.1 Concrete models for the groups in (i)–(iv)

It is useful to describe the groups and maps in the four exact sequences by explicit concrete models.

(i) The case $A \cong \mathbb{Z} \oplus \mathbb{Z}/2$.

The symbol

$$\mathbb{Z} \oplus \mathbb{Z}/2$$

means the direct sum of the two abelian groups. Concretely,

$$A = \mathbb{Z} \oplus \mathbb{Z}/2 = \{(n, \bar{\varepsilon}) \mid n \in \mathbb{Z}, \bar{\varepsilon} \in \mathbb{Z}/2\},$$

with addition defined componentwise:

$$(n, \bar{\varepsilon}) + (m, \bar{\delta}) = (n + m, \overline{\varepsilon + \delta}).$$

Thus elements of A look like

$$(0, \bar{0}), \quad (1, \bar{0}), \quad (5, \bar{1}), \quad (-3, \bar{1}), \dots$$

The first coordinate is any integer, while the second coordinate is either $\bar{0}$ or $\bar{1}$.

A concrete exact sequence is

$$0 \longrightarrow \mathbb{Z}/2 \xrightarrow{\iota} \mathbb{Z} \oplus \mathbb{Z}/2 \xrightarrow{\pi} \mathbb{Z} \longrightarrow 0,$$

where

$$\iota(\bar{\varepsilon}) = (0, \bar{\varepsilon}), \quad \pi(n, \bar{\varepsilon}) = n.$$

Then

$$\text{im}(\iota) = \{(0, \bar{0}), (0, \bar{1})\},$$

and

$$\ker(\pi) = \{(0, \bar{0}), (0, \bar{1})\}.$$

So indeed

$$\ker(\pi) = \text{im}(\iota),$$

and this realizes the exact sequence concretely.

(ii) The two possibilities for A .

In

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0,$$

there are two possible isomorphism types for A .

First possibility: $A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2$.

Here

$$A = \mathbb{Z}/2 \oplus \mathbb{Z}/2 = \{(\bar{a}, \bar{b}) \mid \bar{a}, \bar{b} \in \mathbb{Z}/2\},$$

whose elements are

$$(\bar{0}, \bar{0}), \quad (\bar{1}, \bar{0}), \quad (\bar{0}, \bar{1}), \quad (\bar{1}, \bar{1}).$$

Define

$$\iota(\bar{a}) = (\bar{a}, \bar{0}), \quad \pi(\bar{a}, \bar{b}) = \bar{b}.$$

Then

$$\text{im}(\iota) = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\},$$

and

$$\ker(\pi) = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\}.$$

So this gives the split exact sequence

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Second possibility: $A \cong \mathbb{Z}/4$.

Take

$$0 \longrightarrow \mathbb{Z}/2 \xrightarrow{\iota} \mathbb{Z}/4 \xrightarrow{\pi} \mathbb{Z}/2 \longrightarrow 0,$$

where

$$\iota(\bar{a}) = \bar{2a} \in \mathbb{Z}/4, \quad \pi(\bar{x} \bmod 4) = \bar{x} \bmod 2.$$

Concretely,

$$\mathbb{Z}/4 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}.$$

The image of ι is

$$\text{im}(\iota) = \{\bar{0}, \bar{2}\},$$

while the kernel of π is also

$$\ker(\pi) = \{\bar{0}, \bar{2}\}.$$

Hence

$$\ker(\pi) = \text{im}(\iota),$$

so the sequence is exact.

Thus case (ii) admits both

$$A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \quad \text{and} \quad A \cong \mathbb{Z}/4.$$

(iii) The case $A = 0$ and $\alpha = \times 2$ or $\times(-2)$.

We found that

$$0 \longrightarrow A \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0$$

forces

$$A = 0, \quad \alpha(n) = 2n \text{ or } \alpha(n) = -2n.$$

A concrete realization is

$$0 \longrightarrow 0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Here

$$(\times 2)(n) = 2n.$$

Its image is the subgroup of even integers:

$$2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}.$$

The last map is reduction modulo 2, and therefore

$$\mathbb{Z}/(2\mathbb{Z}) \cong \mathbb{Z}/2.$$

Since multiplication by 2 on $\mathbb{Z}$ has trivial kernel,

$$\ker(\times 2) = 0,$$

the group A is the trivial group:

$$A = 0.$$

(iv) The case $A \cong \mathbb{Z}/6$ and $\alpha = \pm \text{id}_{\mathbb{Z}}$.

We found that

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow 0$$

forces

$$A \cong \mathbb{Z}/6, \quad \alpha = \pm \text{id}_{\mathbb{Z}}.$$

A concrete model for the beginning is

$$0 \longrightarrow \mathbb{Z}/3 \xrightarrow{\iota} \mathbb{Z}/6 \xrightarrow{\pi} \mathbb{Z}/2 \longrightarrow 0,$$

where

$$\iota(\bar{k} \bmod 3) = \overline{2k} \bmod 6, \quad \pi(\bar{x} \bmod 6) = \bar{x} \bmod 2.$$

The image of ι is

$$\text{im}(\iota) = \{\bar{0}, \bar{2}, \bar{4}\} \subset \mathbb{Z}/6.$$

The kernel of π is precisely the even residue classes modulo 6:

$$\ker(\pi) = \{\bar{0}, \bar{2}, \bar{4}\}.$$

Hence

$$\ker(\pi) = \text{im}(\iota),$$

so this part is exact.

After that, the map

$$\mathbb{Z}/2 \longrightarrow \mathbb{Z}$$

must be the zero map, since $\mathbb{Z}$ has no nonzero element of order 2. Therefore the end of the sequence is

$$\mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow 0,$$

with α an isomorphism. Thus

$$\alpha = \text{id}_{\mathbb{Z}} \quad \text{or} \quad \alpha = -\text{id}_{\mathbb{Z}}.$$

So a concrete exact sequence is

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow \mathbb{Z}/6 \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \longrightarrow 0,$$

where the map $\mathbb{Z}/2 \rightarrow \mathbb{Z}$ is the zero homomorphism.

Summary of the concrete models.

$$(i) \quad A \cong \mathbb{Z} \oplus \mathbb{Z}/2,$$

with elements $(n, \bar{\varepsilon})$.

$$(ii) \quad A \cong \mathbb{Z}/4 \quad \text{or} \quad A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

$$(iii) \quad A = 0, \quad \alpha(n) = \pm 2n.$$

$$(iv) \quad A \cong \mathbb{Z}/6, \quad \alpha = \pm \text{id}_{\mathbb{Z}}.$$

These explicit realizations show what the abstract conclusions mean in concrete algebraic terms.

6 Problem 2 A five-lemma type isomorphism criterion

Statement

Consider a commutative diagram

$$\begin{array}{ccccccccc} A_1 & \xrightarrow{f_1} & A_2 & \xrightarrow{f_2} & A_3 & \xrightarrow{f_3} & A_4 & \xrightarrow{f_4} & A_5 \\ h_1 \downarrow & & h_2 \downarrow & & h_3 \downarrow & & h_4 \downarrow & & h_5 \downarrow \\ B_1 & \xrightarrow{g_1} & B_2 & \xrightarrow{g_2} & B_3 & \xrightarrow{g_3} & B_4 & \xrightarrow{g_4} & B_5 \end{array}$$

in which the rows are exact and each square commutes. If h_1, h_2, h_4 , and h_5 are isomorphisms, show that h_3 is also an isomorphism.

Solution.

This is a standard form of the five lemma. We prove that h_3 is injective and surjective.

Injectivity of h_3 . Let $a \in A_3$ satisfy

$$h_3(a) = 0.$$

Apply g_3 to this equality. Since the diagram commutes,

$$g_3(h_3(a)) = h_4(f_3(a)).$$

The left-hand side is zero, so

$$h_4(f_3(a)) = 0.$$

Because h_4 is an isomorphism, it is injective, hence

$$f_3(a) = 0.$$

By exactness of the top row at A_3 ,

$$\ker(f_3) = \operatorname{im}(f_2),$$

so there exists $b \in A_2$ such that

$$a = f_2(b).$$

Now

$$0 = h_3(a) = h_3(f_2(b)) = g_2(h_2(b)),$$

again by commutativity. Hence

$$h_2(b) \in \ker(g_2) = \operatorname{im}(g_1)$$

by exactness of the bottom row. Therefore there exists $c \in B_1$ such that

$$g_1(c) = h_2(b).$$

Since h_1 is an isomorphism, there exists $d \in A_1$ with

$$h_1(d) = c.$$

Then commutativity of the left square gives

$$h_2(f_1(d)) = g_1(h_1(d)) = g_1(c) = h_2(b).$$

Because h_2 is injective, we conclude

$$f_1(d) = b.$$

Thus

$$a = f_2(b) = f_2(f_1(d)) = 0,$$

since consecutive maps in an exact sequence compose to zero. Therefore $\ker(h_3) = 0$, so h_3 is injective.

Surjectivity of h_3 . Let $y \in B_3$. We want to find $x \in A_3$ such that

$$h_3(x) = y.$$

First consider $g_3(y) \in B_4$. Since h_4 is surjective, there exists $a_4 \in A_4$ with

$$h_4(a_4) = g_3(y).$$

Apply g_4 to both sides:

$$g_4(h_4(a_4)) = g_4(g_3(y)) = 0.$$

By commutativity,

$$h_5(f_4(a_4)) = 0.$$

Since h_5 is injective, it follows that

$$f_4(a_4) = 0.$$

By exactness of the top row at A_4 ,

$$a_4 \in \ker(f_4) = \operatorname{im}(f_3),$$

so there exists $x_0 \in A_3$ such that

$$f_3(x_0) = a_4.$$

Now compute:

$$g_3(h_3(x_0)) = h_4(f_3(x_0)) = h_4(a_4) = g_3(y).$$

Hence

$$g_3(y - h_3(x_0)) = 0.$$

So

$$y - h_3(x_0) \in \ker(g_3) = \operatorname{im}(g_2)$$

by exactness of the bottom row. Therefore there exists $b_2 \in B_2$ such that

$$g_2(b_2) = y - h_3(x_0).$$

Since h_2 is surjective, choose $a_2 \in A_2$ with

$$h_2(a_2) = b_2.$$

Set

$$x = x_0 + f_2(a_2) \in A_3.$$

Then

$$h_3(x) = h_3(x_0) + h_3(f_2(a_2)) = h_3(x_0) + g_2(h_2(a_2)) = h_3(x_0) + g_2(b_2) = h_3(x_0) + (y - h_3(x_0)) = y.$$

Thus every $y \in B_3$ lies in the image of h_3 , so h_3 is surjective.

Since h_3 is both injective and surjective, it is an isomorphism. Therefore

h_3 is an isomorphism.

7 Theory, definitions, and examples for Problems 1 and 2

7.1 Abelian groups, homomorphisms, kernels, and images

An *abelian group* is a group $(G, +)$ such that

$$a + b = b + a \quad \text{for all } a, b \in G.$$

Typical examples are

$$\mathbb{Z}, \quad \mathbb{Z}/n\mathbb{Z}, \quad \mathbb{Z} \oplus \mathbb{Z}/2.$$

A *homomorphism* of abelian groups is a map

$$f : G \rightarrow H$$

satisfying

$$f(x + y) = f(x) + f(y) \quad \text{for all } x, y \in G.$$

Two fundamental constructions attached to a homomorphism are the *kernel* and the *image*:

$$\ker(f) = \{x \in G : f(x) = 0\}, \quad \text{im}(f) = \{f(x) : x \in G\}.$$

Recall:

- f is injective if and only if $\ker(f) = 0$,
- f is surjective if and only if $\text{im}(f) = H$,
- f is an isomorphism if and only if it is both injective and surjective.

7.2 Exact sequences

A sequence of group homomorphisms

$$\cdots \longrightarrow A_{i-1} \xrightarrow{f_{i-1}} A_i \xrightarrow{f_i} A_{i+1} \longrightarrow \cdots$$

is said to be *exact at* A_i if

$$\text{im}(f_{i-1}) = \ker(f_i).$$

It is called an *exact sequence* if it is exact at every term.

Thus exactness means that the elements arriving at a group are exactly those elements that are killed by the next map.

A particularly important case is a *short exact sequence*:

$$0 \longrightarrow A \xrightarrow{\iota} B \xrightarrow{\pi} C \longrightarrow 0.$$

Exactness here means:

- ι is injective,

- π is surjective,
- $\text{im}(\iota) = \ker(\pi)$.

So A sits inside B , and C is the quotient of B by that copy of A . In fact,

$$C \cong B/\iota(A).$$

7.3 Direct sums

For abelian groups G and H , the *direct sum*

$$G \oplus H$$

is the set of pairs

$$(g, h), \quad g \in G, \quad h \in H,$$

with componentwise addition:

$$(g, h) + (g', h') = (g + g', h + h').$$

For example,

$$\mathbb{Z} \oplus \mathbb{Z}/2 = \{(n, \bar{\varepsilon}) : n \in \mathbb{Z}, \bar{\varepsilon} \in \mathbb{Z}/2\}.$$

Its elements look like

$$(0, \bar{0}), \quad (0, \bar{1}), \quad (5, \bar{0}), \quad (5, \bar{1}), \quad (-3, \bar{1}), \dots$$

and addition is

$$(a, \bar{\varepsilon}) + (b, \bar{\delta}) = (a + b, \overline{\varepsilon + \delta}).$$

It is important that the second coordinate is *independent* of the first one. Thus

$$(4, \bar{0}) \neq (4, \bar{1}).$$

The symbol $\mathbb{Z}/2$ in the second coordinate does not merely record the parity of the integer in the first coordinate; it is an additional independent order-two component.

7.4 Split short exact sequences

A short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

is said to *split* if

$$B \cong A \oplus C.$$

This means that the middle group decomposes as two independent parts.

A key fact used in Problem 1 is that every short exact sequence

$$0 \longrightarrow K \longrightarrow B \longrightarrow \mathbb{Z} \longrightarrow 0$$

splits. The reason is that $\mathbb{Z}$ is a free abelian group.

A direct proof is very concrete. Suppose

$$0 \longrightarrow K \longrightarrow B \xrightarrow{p} \mathbb{Z} \longrightarrow 0$$

is exact. Since p is surjective, choose $b \in B$ such that

$$p(b) = 1.$$

Define

$$s : \mathbb{Z} \longrightarrow B, \quad s(n) = nb.$$

Then

$$p(s(n)) = p(nb) = np(b) = n,$$

so $p \circ s = \text{id}_{\mathbb{Z}}$. Hence s is a right inverse of p , and the sequence splits. Therefore

$$B \cong K \oplus \mathbb{Z}.$$

7.5 Homomorphisms $\mathbb{Z} \rightarrow \mathbb{Z}$

Every homomorphism

$$\alpha : \mathbb{Z} \rightarrow \mathbb{Z}$$

is determined by the image of 1. If $\alpha(1) = n$, then for every $k \in \mathbb{Z}$,

$$\alpha(k) = nk.$$

So every such homomorphism is multiplication by some integer:

$$\alpha = \times n.$$

Its image is

$$\text{im}(\alpha) = n\mathbb{Z},$$

and its cokernel is

$$\text{coker}(\alpha) \cong \mathbb{Z}/n\mathbb{Z}$$

for $n \neq 0$. More precisely, if $n \neq 0$, then

$$\mathbb{Z}/\text{im}(\alpha) = \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/|n|.$$

Also:

$$\ker(\times n) = 0 \quad \text{for } n \neq 0.$$

Thus multiplication by 2 or -2 on $\mathbb{Z}$ is injective.

7.6 Homomorphisms from finite groups to $\mathbb{Z}$

If G is a finite abelian group and

$$f : G \rightarrow \mathbb{Z}$$

is a homomorphism, then $f = 0$.

Indeed, let $x \in G$. Since G is finite, x has finite order, say $mx = 0$ for some $m > 0$. Applying f gives

$$mf(x) = f(mx) = f(0) = 0.$$

But the only element of $\mathbb{Z}$ annihilated by a nonzero integer is 0. Hence $f(x) = 0$. Since this holds for all $x \in G$, the map is zero.

In particular,

$$\text{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0.$$

This fact is essential in part (iv) of Problem 1.

7.7 Small finite abelian groups

Problem 1 also uses the classification of small abelian groups.

Groups of order 4. Up to isomorphism, there are exactly two abelian groups of order 4:

$$\mathbb{Z}/4, \quad \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Groups of order 6. Every abelian group of order 6 is cyclic, hence isomorphic to

$$\mathbb{Z}/6.$$

Equivalently, by the Chinese remainder theorem,

$$\mathbb{Z}/6 \cong \mathbb{Z}/3 \oplus \mathbb{Z}/2.$$

7.8 Problem 1: determining A and α

We now revisit the four exact sequences of Problem 1.

(i) **The sequence** $0 \rightarrow \mathbb{Z}/2 \rightarrow A \rightarrow \mathbb{Z} \rightarrow 0$

Consider

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z} \longrightarrow 0.$$

Since the sequence ends in $\mathbb{Z}$, it splits. Therefore

$$A \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

A concrete model is

$$0 \longrightarrow \mathbb{Z}/2 \xrightarrow{\iota} \mathbb{Z} \oplus \mathbb{Z}/2 \xrightarrow{\pi} \mathbb{Z} \longrightarrow 0,$$

where

$$\iota(\bar{\varepsilon}) = (0, \bar{\varepsilon}), \quad \pi(n, \bar{\varepsilon}) = n.$$

Then

$$\text{im}(\iota) = \{(0, \bar{0}), (0, \bar{1})\},$$

while

$$\ker(\pi) = \{(0, \bar{0}), (0, \bar{1})\}.$$

Hence the sequence is exact.

So in this case,

$$\boxed{A \cong \mathbb{Z} \oplus \mathbb{Z}/2.}$$

(ii) The sequence $0 \rightarrow \mathbb{Z}/2 \rightarrow A \rightarrow \mathbb{Z}/2 \rightarrow 0$

Now consider

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Since both outer groups have order 2, the middle group A must have order 4. Hence

$$A \cong \mathbb{Z}/4 \quad \text{or} \quad A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Both possibilities actually occur.

Split example. Take

$$A = \mathbb{Z}/2 \oplus \mathbb{Z}/2.$$

Define

$$\iota(\bar{a}) = (\bar{a}, \bar{0}), \quad \pi(\bar{a}, \bar{b}) = \bar{b}.$$

Then

$$\text{im}(\iota) = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\}$$

and

$$\ker(\pi) = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\}.$$

So this gives the split exact sequence

$$0 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/2 \oplus \mathbb{Z}/2 \rightarrow \mathbb{Z}/2 \rightarrow 0.$$

Non-split example. Take

$$A = \mathbb{Z}/4.$$

Define

$$\iota(\bar{a}) = \bar{2a} \in \mathbb{Z}/4, \quad \pi(\bar{x} \bmod 4) = \bar{x} \bmod 2.$$

Then

$$\text{im}(\iota) = \{\bar{0}, \bar{2}\}$$

and

$$\ker(\pi) = \{\bar{0}, \bar{2}\}.$$

So this gives another exact sequence

$$0 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z}/4 \rightarrow \mathbb{Z}/2 \rightarrow 0.$$

Thus the exact sequence does not determine A uniquely:

$$\boxed{A \cong \mathbb{Z}/4 \quad \text{or} \quad A \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2.}$$

(iii) The sequence $0 \rightarrow A \rightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$

Consider

$$0 \longrightarrow A \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Since every homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$ is multiplication by an integer, write

$$\alpha = \times n.$$

Its cokernel is

$$\text{coker}(\alpha) \cong \mathbb{Z}/n\mathbb{Z}.$$

Exactness tells us that this cokernel is $\mathbb{Z}/2$, so necessarily

$$|n| = 2.$$

Hence

$$\alpha = \times 2 \quad \text{or} \quad \alpha = \times (-2).$$

Now

$$A = \ker(\alpha).$$

But multiplication by ± 2 on $\mathbb{Z}$ has trivial kernel, so

$$A = 0.$$

A concrete model is

$$0 \longrightarrow 0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

The map $\mathbb{Z} \rightarrow \mathbb{Z}/2$ is reduction modulo 2, and the image of $\times 2$ is the subgroup of even integers.

Thus

$$\boxed{A = 0, \quad \alpha = \times 2 \text{ or } \times (-2).}$$

(iv) The sequence $0 \rightarrow \mathbb{Z}/3 \rightarrow A \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \rightarrow 0$

Now consider

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} \mathbb{Z} \longrightarrow 0.$$

First, the map

$$\mathbb{Z}/2 \rightarrow \mathbb{Z}$$

must be zero, since every homomorphism from a finite group to $\mathbb{Z}$ is zero.

Therefore exactness at the copy of $\mathbb{Z}$ before α gives

$$\ker(\alpha) = \text{im}(\mathbb{Z}/2 \rightarrow \mathbb{Z}) = 0.$$

So α is injective.

Exactness at the last $\mathbb{Z}$ shows that α is surjective as well. Hence α is an automorphism of $\mathbb{Z}$, so

$$\alpha = \pm \text{id}_{\mathbb{Z}}.$$

Now the earlier part reduces to

$$0 \longrightarrow \mathbb{Z}/3 \longrightarrow A \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Thus A is an abelian group of order 6, and therefore

$$A \cong \mathbb{Z}/6.$$

A concrete model is

$$0 \longrightarrow \mathbb{Z}/3 \xrightarrow{\iota} \mathbb{Z}/6 \xrightarrow{\pi} \mathbb{Z}/2 \longrightarrow 0,$$

where

$$\iota(\bar{k} \bmod 3) = \overline{2k} \bmod 6, \quad \pi(\bar{x} \bmod 6) = \bar{x} \bmod 2.$$

Then

$$\text{im}(\iota) = \{\bar{0}, \bar{2}, \bar{4}\}$$

and

$$\ker(\pi) = \{\bar{0}, \bar{2}, \bar{4}\},$$

so the sequence is exact.

Hence

$$\boxed{A \cong \mathbb{Z}/6, \quad \alpha = \pm \text{id}_{\mathbb{Z}}.}$$

7.9 Summary of Problem 1

The four answers are:

$$\boxed{\text{(i) } A \cong \mathbb{Z} \oplus \mathbb{Z}/2,}$$

$$\boxed{\text{(ii) } A \cong \mathbb{Z}/4 \text{ or } \mathbb{Z}/2 \oplus \mathbb{Z}/2,}$$

$$\boxed{\text{(iii) } A = 0, \alpha = \times 2 \text{ or } \times (-2),}$$

$$\boxed{\text{(iv) } A \cong \mathbb{Z}/6, \alpha = \pm \text{id}_{\mathbb{Z}}.}$$

The main ideas behind these computations are:

- exactness means image equals kernel,
- short exact sequences ending in $\mathbb{Z}$ split,
- every map $\mathbb{Z} \rightarrow \mathbb{Z}$ is multiplication by an integer,
- every map from a finite group to $\mathbb{Z}$ is zero,
- small finite abelian groups can be classified explicitly.

7.10 Commutative diagrams

Problem 2 concerns a commutative diagram

$$\begin{array}{ccccccccc}
 A_1 & \xrightarrow{f_1} & A_2 & \xrightarrow{f_2} & A_3 & \xrightarrow{f_3} & A_4 & \xrightarrow{f_4} & A_5 \\
 h_1 \downarrow & & h_2 \downarrow & & h_3 \downarrow & & h_4 \downarrow & & h_5 \downarrow \\
 B_1 & \xrightarrow{g_1} & B_2 & \xrightarrow{g_2} & B_3 & \xrightarrow{g_3} & B_4 & \xrightarrow{g_4} & B_5.
 \end{array}$$

A diagram is said to *commute* if every square commutes. For example, the square involving A_2, A_3, B_2, B_3 commutes if

$$h_3 \circ f_2 = g_2 \circ h_2.$$

That means that going right and then down gives the same result as going down and then right.

7.11 The five lemma

The theorem used in Problem 2 is a classical result called the *five lemma*. One standard form says:

Suppose we have a commutative diagram with exact rows

$$\begin{array}{ccccccccc}
 A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\
 h_1 \downarrow & & h_2 \downarrow & & h_3 \downarrow & & h_4 \downarrow & & h_5 \downarrow \\
 B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5
 \end{array}$$

and suppose h_1, h_2, h_4, h_5 are isomorphisms. Then h_3 is also an isomorphism.

The theorem is proved by *diagram chasing*: one follows an element through the diagram, uses exactness to rewrite kernels as images, and uses the isomorphisms in the neighboring positions to lift and compare elements.

7.12 Why the five lemma works

To show that h_3 is an isomorphism, it suffices to prove:

- h_3 is injective,
- h_3 is surjective.

Injectivity idea. Suppose $a \in A_3$ and

$$h_3(a) = 0.$$

Then one applies the next map to obtain information about $f_3(a)$, uses the injectivity of h_4 , and concludes

$$f_3(a) = 0.$$

By exactness of the top row, a must come from A_2 , say

$$a = f_2(b).$$

Then one pushes b down to B_2 , uses exactness of the bottom row and the isomorphisms h_1, h_2 , and concludes that b actually comes from A_1 . Hence

$$a = f_2(f_1(\cdot)) = 0.$$

So h_3 is injective.

Surjectivity idea. Now take $y \in B_3$. One wants to produce $x \in A_3$ such that

$$h_3(x) = y.$$

Push y forward to B_4 , lift it back to A_4 using the isomorphism h_4 , and then use exactness of the top row to obtain a partial preimage $x_0 \in A_3$. The difference

$$y - h_3(x_0)$$

is then killed by g_3 , so by exactness of the bottom row it comes from B_2 . Using the isomorphism h_2 , lift that correction back to A_2 , adjust x_0 by something from $\text{im}(f_2)$, and obtain the desired x . So h_3 is surjective.

Hence h_3 is an isomorphism.

7.13 Problem 2 as a concrete theorem

Problem 2 is therefore a direct instance of the five lemma:

If the rows are exact, the squares commute, and h_1, h_2, h_4, h_5 are isomorphisms, then h_3 is an isomorphism.

This result is fundamental in algebra, topology, and homological algebra, because it allows us to compare complicated exact sequences by comparing their neighboring terms.

7.14 A simple example illustrating the five lemma

Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Now place two copies of it in a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\times 2} & \mathbb{Z} & \longrightarrow & \mathbb{Z}/2 \longrightarrow 0 \\ & & \downarrow \text{id} & & \downarrow h_3 & & \downarrow \text{id} \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\times 2} & \mathbb{Z} & \longrightarrow & \mathbb{Z}/2 \longrightarrow 0. \end{array}$$

Since the two outside vertical maps are identities and the rows are exact, commutativity forces the middle map h_3 to be an isomorphism.

This simple example shows the general principle: if exact rows agree on the neighboring terms, then the middle term is forced to agree as well.

7.15 Conceptual summary of Problems 1 and 2

Problem 1 is about the internal structure of groups appearing in exact sequences. It uses:

- kernels and images,
- short exact sequences,
- direct sums,
- classification of small abelian groups,
- explicit homomorphisms $\mathbb{Z} \rightarrow \mathbb{Z}$.

Problem 2 is about comparing two exact sequences through a commutative diagram. It uses:

- commutative diagrams,
- exactness in both rows,
- injectivity and surjectivity,
- diagram chasing,
- the five lemma.

Together, these two problems illustrate two central themes of homological algebra:

1. exactness determines the possible algebraic structure of unknown groups and maps;
2. commutative exact diagrams allow us to transfer information from known terms to unknown ones.

8 Problem 3 Suspension of a simplicial complex and induced maps on homology

Statement

Let K be a simplicial complex in $\mathbb{R}^m$, and consider this as lying inside $\mathbb{R}^{m+1}$ as the vectors of the form

$$(x_1, \dots, x_m, 0).$$

Let

$$e_+ = (0, \dots, 0, 1) \in \mathbb{R}^{m+1}, \quad e_- = (0, \dots, 0, -1) \in \mathbb{R}^{m+1}.$$

The *suspension* of K is the simplicial complex in $\mathbb{R}^{m+1}$

$$SK := K \cup \{\langle v_0, \dots, v_n, e_+ \rangle, \langle v_0, \dots, v_n, e_- \rangle \mid \langle v_0, \dots, v_n \rangle \in K\}.$$

- (a) Show that SK is a simplicial complex, and that if $|K| \cong S^n$ then $|SK| \cong S^{n+1}$.
- (b) Using the Mayer–Vietoris sequence, show that if K is connected then

$$H_0(SK) \cong \mathbb{Z}, \quad H_1(SK) = 0, \quad H_i(SK) \cong H_{i-1}(K) \text{ if } i \geq 2.$$

- (c) If $f : K \rightarrow K$ is a simplicial map, let $Sf : SK \rightarrow SK$ be the unique simplicial map which agrees with f on the subcomplex K and fixes the points e_+ and e_- . Show that under the isomorphism in (b), the maps f_* and $(Sf)_*$ agree.
- (d) Deduce that for every $n \geq 1$ and $d \in \mathbb{Z}$ there is a map

$$f : S^n \rightarrow S^n$$

so that f_* induces multiplication by d on

$$H_n(S^n) \cong \mathbb{Z}.$$

Solution.

(a) The suspension SK is a simplicial complex.

Let

$$\sigma = \langle v_0, \dots, v_n \rangle \in K.$$

Then the added simplices are the cones

$$\langle v_0, \dots, v_n, e_+ \rangle, \quad \langle v_0, \dots, v_n, e_- \rangle.$$

To check that SK is a simplicial complex, we verify the two defining properties.

First, every face of a simplex in SK again lies in SK . Indeed, if a simplex already lies in K , then all its faces lie in K because K is a simplicial complex. If the simplex is of the form

$$\langle v_0, \dots, v_n, e_+ \rangle \quad \text{or} \quad \langle v_0, \dots, v_n, e_- \rangle,$$

then any face is either:

- a face $\langle v_{i_0}, \dots, v_{i_r} \rangle$ of σ , hence in $K \subset SK$, or
- a simplex of the form $\langle v_{i_0}, \dots, v_{i_r}, e_+ \rangle$ or $\langle v_{i_0}, \dots, v_{i_r}, e_- \rangle$, again by construction in SK .

Second, the intersection of two simplices of SK is either empty or a common face. This is clear if both lie in K , since K is a simplicial complex. If both contain e_+ , then they are cones on simplices of K , and their intersection is the cone on the intersection of the base simplices, hence a common face. The same argument applies to two simplices containing e_- . If one simplex contains e_+ and the other contains e_- , then they cannot share either e_+ or e_- , so their intersection is exactly the intersection of the corresponding base simplices in K , again a common face or empty.

Thus SK is a simplicial complex.

Now suppose $|K| \cong S^n$. Geometrically, $|SK|$ is obtained by taking two cones on $|K|$, one with apex e_+ and one with apex e_- , and gluing them along the common base $|K|$. This is exactly the topological suspension of $|K|$:

$$|SK| \cong \Sigma|K|.$$

Since the suspension of S^n is S^{n+1} , we obtain

$$|SK| \cong \Sigma S^n \cong S^{n+1}.$$

Hence

$$\boxed{|K| \cong S^n \implies |SK| \cong S^{n+1}.}$$

(b) Homology of the suspension.

Let

$$U = \text{the cone on } K \text{ with apex } e_+, \quad V = \text{the cone on } K \text{ with apex } e_-.$$

Then

$$SK = U \cup V, \quad U \cap V = K.$$

Both U and V are cones, hence contractible. Therefore

$$H_i(U) = H_i(V) = 0 \quad (i \geq 1),$$

and

$$H_0(U) \cong \mathbb{Z}, \quad H_0(V) \cong \mathbb{Z}.$$

Apply the Mayer–Vietoris sequence to the decomposition $SK = U \cup V$:

$$\cdots \rightarrow H_i(K) \rightarrow H_i(U) \oplus H_i(V) \rightarrow H_i(SK) \xrightarrow{\delta} H_{i-1}(K) \rightarrow H_{i-1}(U) \oplus H_{i-1}(V) \rightarrow \cdots$$

If $i \geq 2$, then

$$H_i(U) \oplus H_i(V) = 0 \quad \text{and} \quad H_{i-1}(U) \oplus H_{i-1}(V) = 0,$$

so the exact sequence becomes

$$0 \rightarrow H_i(SK) \xrightarrow{\delta} H_{i-1}(K) \rightarrow 0.$$

Thus

$$\boxed{H_i(SK) \cong H_{i-1}(K) \quad (i \geq 2).}$$

Now consider the low-dimensional part:

$$0 \rightarrow H_1(SK) \rightarrow H_0(K) \xrightarrow{\phi} H_0(U) \oplus H_0(V) \rightarrow H_0(SK) \rightarrow 0.$$

Since K is connected,

$$H_0(K) \cong \mathbb{Z}.$$

Also

$$H_0(U) \oplus H_0(V) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

The map ϕ is induced by the inclusions $K \hookrightarrow U$ and $K \hookrightarrow V$, so on generators it is the diagonal map

$$\phi(1) = (1, 1).$$

Hence ϕ is injective, so

$$H_1(SK) = 0.$$

Its cokernel is

$$(\mathbb{Z} \oplus \mathbb{Z}) / \Delta(\mathbb{Z}) \cong \mathbb{Z},$$

so

$$H_0(SK) \cong \mathbb{Z}.$$

Therefore

$$\boxed{H_0(SK) \cong \mathbb{Z}, \quad H_1(SK) = 0, \quad H_i(SK) \cong H_{i-1}(K) \text{ for } i \geq 2.}$$

(c) Compatibility of f_* and $(Sf)_*$.

Let $f : K \rightarrow K$ be simplicial, and let $Sf : SK \rightarrow SK$ be its suspension, fixing e_+ and e_- . Since Sf fixes the two cone points, it preserves the subcomplexes U and V , and on

$$U \cap V = K$$

it restricts exactly to f . Thus we get a morphism between the Mayer–Vietoris sequences of the decomposition

$$SK = U \cup V.$$

For $i \geq 2$, the isomorphism

$$H_i(SK) \xrightarrow{\cong} H_{i-1}(K)$$

from part (b) is precisely the connecting homomorphism δ in Mayer–Vietoris. By naturality of the Mayer–Vietoris sequence, the diagram

$$\begin{array}{ccc} H_i(SK) & \xrightarrow{\delta} & H_{i-1}(K) \\ (Sf)_* \downarrow & & \downarrow f_* \\ H_i(SK) & \xrightarrow{\delta} & H_{i-1}(K) \end{array}$$

commutes.

Since δ is an isomorphism, under the identification

$$H_i(SK) \cong H_{i-1}(K),$$

the map $(Sf)_*$ corresponds exactly to f_* . That is,

$$\boxed{\delta \circ (Sf)_* = f_* \circ \delta.}$$

(d) Maps of arbitrary degree on spheres.

We prove by induction on $n \geq 1$ that for every integer d there exists a map

$$f : S^n \rightarrow S^n$$

whose induced map on $H_n(S^n) \cong \mathbb{Z}$ is multiplication by d .

For $n = 1$, consider the map

$$f_d : S^1 \rightarrow S^1, \quad f_d(z) = z^d \quad (z \in S^1 \subset \mathbb{C}).$$

This is a continuous map of degree d , so

$$(f_d)_* : H_1(S^1) \rightarrow H_1(S^1)$$

is multiplication by d .

Now assume the statement holds for S^n . Then there is a map

$$f : S^n \rightarrow S^n$$

with f_* equal to multiplication by d on $H_n(S^n)$. Suspend f to obtain

$$Sf : S^{n+1} \rightarrow S^{n+1},$$

using part (a), since $S^{n+1} \cong \Sigma S^n$. By part (c), under the isomorphism

$$H_{n+1}(S^{n+1}) \cong H_n(S^n),$$

the map $(Sf)_*$ agrees with f_* . Hence $(Sf)_*$ is also multiplication by d .

Therefore, by induction,

for every $n \geq 1$ and $d \in \mathbb{Z}$, there exists $f : S^n \rightarrow S^n$ with $f_* = \times d$ on $H_n(S^n) \cong \mathbb{Z}$.

9 Problem 4 Rational homology detects only the free part

Statement

If K is a simplicial complex with

$$H_i(K) \cong \mathbb{Z}^r \oplus F,$$

where F is a finite abelian group, show that

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^r.$$

Hint: There is a chain map $C_\bullet(K) \rightarrow C_\bullet(K; \mathbb{Q})$.

Solution.

Recall that the simplicial chain groups with rational coefficients are obtained by tensoring the integral chain groups with $\mathbb{Q}$:

$$C_n(K; \mathbb{Q}) = C_n(K) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

The boundary maps extend by linearity, so the whole rational chain complex is

$$C_\bullet(K; \mathbb{Q}) = C_\bullet(K) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Now $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module, so tensoring with $\mathbb{Q}$ preserves exact sequences. Therefore homology commutes with tensoring:

$$H_i(K; \mathbb{Q}) \cong H_i(K) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Using the given decomposition

$$H_i(K) \cong \mathbb{Z}^r \oplus F,$$

we obtain

$$H_i(K; \mathbb{Q}) \cong (\mathbb{Z}^r \oplus F) \otimes_{\mathbb{Z}} \mathbb{Q} \cong (\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Q}) \oplus (F \otimes_{\mathbb{Z}} \mathbb{Q}).$$

We now compute the two pieces.

First,

$$\mathbb{Z}^r \otimes_{\mathbb{Z}} \mathbb{Q} \cong \mathbb{Q}^r.$$

Second, if F is finite, then

$$F \otimes_{\mathbb{Z}} \mathbb{Q} = 0.$$

Indeed, for any $x \in F$, there exists $m > 0$ such that $mx = 0$. Then for any $q \in \mathbb{Q}$,

$$x \otimes q = mx \otimes \frac{q}{m} = 0 \otimes \frac{q}{m} = 0.$$

So every pure tensor is zero, hence the whole tensor product is zero.

Combining the two computations gives

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^r \oplus 0 \cong \mathbb{Q}^r.$$

Therefore

$$\boxed{H_i(K; \mathbb{Q}) \cong \mathbb{Q}^r.}$$

In particular, rational homology remembers only the free part of $H_i(K)$ and kills all torsion. For example:

- if

$$H_i(K) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/6,$$

then

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^2;$$

- if

$$H_i(K) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/9,$$

then

$$H_i(K; \mathbb{Q}) \cong 0;$$

- if

$$H_i(K) \cong \mathbb{Z}^3,$$

then

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^3.$$

9.1 Concrete examples of suspensions of simplicial complexes

It is very helpful to see the suspension construction on explicit simplicial complexes with concrete vertices and simplices. Recall that if $K \subset \mathbb{R}^m$ is a simplicial complex, then we regard it inside $\mathbb{R}^{m+1}$ by sending

$$(x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0),$$

and then add the two new vertices

$$e_+ = (0, \dots, 0, 1), \quad e_- = (0, \dots, 0, -1).$$

For each simplex

$$\sigma = \langle v_0, \dots, v_n \rangle \in K,$$

the suspension SK contains the two cones

$$\langle v_0, \dots, v_n, e_+ \rangle, \quad \langle v_0, \dots, v_n, e_- \rangle.$$

Geometrically, one keeps the old complex K in the middle hyperplane, adds a “north pole” e_+ , adds a “south pole” e_- , and joins every simplex of K to both poles.

Example 1: K is two points (S^0)

Let $K \subset \mathbb{R}$ have vertices

$$v_0 = (-1), \quad v_1 = (1),$$

and no 1-simplex between them. Thus K consists of the two 0-simplices

$$\langle v_0 \rangle, \quad \langle v_1 \rangle.$$

Topologically,

$$|K| \cong S^0.$$

Now embed K into $\mathbb{R}^2$:

$$v_0 = (-1, 0), \quad v_1 = (1, 0), \quad e_+ = (0, 1), \quad e_- = (0, -1).$$

Then SK has the four vertices

$$v_0, v_1, e_+, e_-,$$

and the four edges

$$\langle v_0, e_+ \rangle, \quad \langle v_1, e_+ \rangle, \quad \langle v_0, e_- \rangle, \quad \langle v_1, e_- \rangle.$$

There are no 2-simplices, since K had no 1-simplices.

Thus $|SK|$ is a loop with four edges:

$$v_0 \rightarrow e_+ \rightarrow v_1 \rightarrow e_- \rightarrow v_0.$$

So

$$|SK| \cong S^1.$$

This is the simplest nontrivial suspension:

$$\Sigma S^0 \cong S^1.$$

This example also shows why connectedness is needed in Problem 3(b). Indeed, K is not connected, and

$$H_0(K) \cong \mathbb{Z} \oplus \mathbb{Z},$$

while

$$H_1(SK) \cong H_1(S^1) \cong \mathbb{Z},$$

so the conclusion $H_1(SK) = 0$ does not hold.

Example 2: K is one edge

Let $K \subset \mathbb{R}$ have vertices

$$v_0 = (-1), \quad v_1 = (1),$$

and one 1-simplex

$$\langle v_0, v_1 \rangle.$$

Thus K is a single interval, so $|K|$ is contractible.

Embed it into $\mathbb{R}^2$:

$$v_0 = (-1, 0), \quad v_1 = (1, 0), \quad e_+ = (0, 1), \quad e_- = (0, -1).$$

Then SK contains:

- the original edge $\langle v_0, v_1 \rangle$,
- the triangle $\langle v_0, v_1, e_+ \rangle$,
- the triangle $\langle v_0, v_1, e_- \rangle$.

Geometrically, $|SK|$ is two filled triangles glued along the common base edge $\langle v_0, v_1 \rangle$. This is a filled rhombus, hence contractible. Therefore

$$H_0(SK) \cong \mathbb{Z}, \quad H_i(SK) = 0 \quad (i \geq 1).$$

This agrees with the homology formula in Problem 3(b): since K is connected and contractible,

$$H_i(K) = 0 \quad (i \geq 1),$$

so

$$H_1(SK) = 0, \quad H_i(SK) \cong H_{i-1}(K) = 0 \quad (i \geq 2).$$

Example 3: K is the boundary of a triangle (S^1)

Let $K \subset \mathbb{R}^2$ have vertices

$$v_0 = (1, 0), \quad v_1 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad v_2 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right),$$

and let K contain the three edges

$$\langle v_0, v_1 \rangle, \quad \langle v_1, v_2 \rangle, \quad \langle v_2, v_0 \rangle,$$

but not the 2-simplex $\langle v_0, v_1, v_2 \rangle$. Thus $|K|$ is the boundary of a triangle, so

$$|K| \cong S^1.$$

Embed K into $\mathbb{R}^3$:

$$v_0 = (1, 0, 0), \quad v_1 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad v_2 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right),$$

and add the two poles

$$e_+ = (0, 0, 1), \quad e_- = (0, 0, -1).$$

Then SK has:

- vertices

$$v_0, v_1, v_2, e_+, e_-;$$

- equatorial edges

$$\langle v_0, v_1 \rangle, \quad \langle v_1, v_2 \rangle, \quad \langle v_2, v_0 \rangle;$$

- edges from the poles to the equator

$$\langle v_i, e_+ \rangle, \quad \langle v_i, e_- \rangle \quad (i = 0, 1, 2);$$

- top 2-simplices

$$\langle v_0, v_1, e_+ \rangle, \quad \langle v_1, v_2, e_+ \rangle, \quad \langle v_2, v_0, e_+ \rangle;$$

- bottom 2-simplices

$$\langle v_0, v_1, e_- \rangle, \quad \langle v_1, v_2, e_- \rangle, \quad \langle v_2, v_0, e_- \rangle.$$

Geometrically this is a triangular bipyramid. Topologically it is a 2-sphere:

$$|SK| \cong S^2.$$

Hence

$$H_0(SK) \cong \mathbb{Z}, \quad H_1(SK) = 0, \quad H_2(SK) \cong \mathbb{Z}.$$

This matches Problem 3(b), since

$$H_1(K) \cong \mathbb{Z},$$

and therefore

$$H_2(SK) \cong H_1(K) \cong \mathbb{Z}.$$

Example 4: K is a filled triangle

Now take the same vertices

$$v_0 = (1, 0), \quad v_1 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad v_2 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right),$$

but this time let K contain not only the boundary edges but also the 2-simplex

$$\langle v_0, v_1, v_2 \rangle.$$

Thus $|K|$ is the whole filled triangle, so K is contractible.

Embed in $\mathbb{R}^3$ as before and add

$$e_+ = (0, 0, 1), \quad e_- = (0, 0, -1).$$

Then SK contains:

- the original filled triangle $\langle v_0, v_1, v_2 \rangle$,
- the tetrahedron $\langle v_0, v_1, v_2, e_+ \rangle$,
- the tetrahedron $\langle v_0, v_1, v_2, e_- \rangle$.

So $|SK|$ is two tetrahedra glued along their common triangular face. This is a 3-ball, hence contractible. Therefore

$$H_0(SK) \cong \mathbb{Z}, \quad H_i(SK) = 0 \quad (i \geq 1).$$

Again this matches Problem 3(b), since K is connected and contractible.

Example 5: K is the boundary of a tetrahedron (S^2)

Let $K \subset \mathbb{R}^3$ be the boundary of a tetrahedron with vertices

$$v_0 = (1, 1, 1), \quad v_1 = (1, -1, -1), \quad v_2 = (-1, 1, -1), \quad v_3 = (-1, -1, 1).$$

Let K contain all vertices, all edges, and all triangular faces of the tetrahedron, but not the interior 3-simplex. Then

$$|K| \cong S^2.$$

Now regard K inside $\mathbb{R}^4$ as lying in the hyperplane $x_4 = 0$, and add

$$e_+ = (0, 0, 0, 1), \quad e_- = (0, 0, 0, -1).$$

Then SK is obtained by joining every simplex of the tetrahedral boundary to both poles. Topologically this is the suspension of S^2 , hence

$$|SK| \cong S^3.$$

Its homology is therefore

$$H_0(SK) \cong \mathbb{Z}, \quad H_1(SK) = 0, \quad H_2(SK) = 0, \quad H_3(SK) \cong \mathbb{Z}.$$

This is the next stage of the pattern

$$\Sigma S^2 \cong S^3.$$

A Mayer–Vietoris picture in Example 3

The example $K = \partial\Delta^2$ makes the Mayer–Vietoris argument very concrete. Let

$$U = \text{the cone on } K \text{ with apex } e_+, \quad V = \text{the cone on } K \text{ with apex } e_-.$$

Then

$$SK = U \cup V, \quad U \cap V = K.$$

Both U and V are contractible, while

$$K \cong S^1.$$

So the Mayer–Vietoris sequence gives

$$H_2(SK) \cong H_1(K) \cong \mathbb{Z}.$$

Geometrically, the one essential loop in the equator K becomes one essential 2-dimensional sphere class after suspension.

A concrete example for part (c)

Take again $K = \partial\Delta^2$, the boundary of a triangle, and define a simplicial automorphism

$$f : K \rightarrow K$$

by cyclically permuting the vertices:

$$f(v_0) = v_1, \quad f(v_1) = v_2, \quad f(v_2) = v_0.$$

Geometrically this is a rotation of the triangle, so on

$$H_1(K) \cong \mathbb{Z}$$

it induces the identity map:

$$f_* = \text{id}.$$

Its suspension

$$Sf : SK \rightarrow SK$$

fixes the poles e_+ and e_- and rotates the equatorial vertices in the same cyclic order. Thus on the suspended sphere $SK \cong S^2$, the map Sf is a rotation of the triangular bipyramid, and therefore on

$$H_2(SK) \cong \mathbb{Z}$$

it also induces the identity. This is a concrete illustration of Problem 3(c): under the isomorphism

$$H_2(SK) \cong H_1(K),$$

the maps f_* and $(Sf)_*$ agree.

A concrete example for part (d)

For the circle $S^1 \subset \mathbb{C}$, define

$$f_d : S^1 \rightarrow S^1, \quad f_d(z) = z^d.$$

This map has degree d , so

$$(f_d)_* : H_1(S^1) \rightarrow H_1(S^1)$$

is multiplication by d .

Suspending this map gives

$$Sf_d : S^2 \rightarrow S^2,$$

which induces multiplication by d on

$$H_2(S^2) \cong \mathbb{Z}.$$

Suspending again gives a map

$$S^3 \rightarrow S^3$$

of the same degree d , and so on. Thus Problem 3(d) says that once one knows how to build degree- d maps on S^1 , suspension produces degree- d maps on all higher spheres.

Summary of the examples

These concrete examples show the basic pattern very clearly:

- suspension of two points gives a circle:

$$\Sigma S^0 \cong S^1;$$

- suspension of an interval is contractible;
- suspension of the boundary of a triangle gives a 2-sphere:

$$\Sigma S^1 \cong S^2;$$

- suspension of a filled triangle is contractible;
- suspension of the boundary of a tetrahedron gives a 3-sphere:

$$\Sigma S^2 \cong S^3.$$

Thus the suspension construction increases sphere dimension by one, while suspending a contractible complex remains contractible.

9.2 Concrete examples for rational homology and the free part

Problem 4 says that if

$$H_i(K) \cong \mathbb{Z}^r \oplus F,$$

where F is a finite abelian group, then

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^r.$$

Thus rational homology keeps the free part and discards the torsion part. The following examples make this concrete.

Example 1: the circle S^1

Let K be the boundary of a triangle. Then $|K| \cong S^1$, and

$$H_1(K) \cong \mathbb{Z}.$$

Here the homology group has one free generator and no torsion. Therefore

$$H_1(K; \mathbb{Q}) \cong \mathbb{Q}.$$

So the one essential loop of the circle remains visible over $\mathbb{Q}$.

Example 2: the 2-sphere S^2

Let K be the boundary of a tetrahedron. Then $|K| \cong S^2$, and

$$H_2(K) \cong \mathbb{Z}.$$

Hence

$$H_2(K; \mathbb{Q}) \cong \mathbb{Q}.$$

Thus the top-dimensional fundamental class of the sphere survives over $\mathbb{Q}$.

Example 3: the torus T^2

Let K be a triangulation of the torus. Then

$$H_1(K) \cong \mathbb{Z}^2, \quad H_2(K) \cong \mathbb{Z}.$$

Therefore

$$H_1(K; \mathbb{Q}) \cong \mathbb{Q}^2, \quad H_2(K; \mathbb{Q}) \cong \mathbb{Q}.$$

So over $\mathbb{Q}$, the torus still has two independent 1-dimensional classes and one 2-dimensional class. Nothing disappears because there is no torsion in these homology groups.

Example 4: the wedge $S^1 \vee S^1$

Let K be a simplicial model of the wedge of two circles. Then

$$H_1(K) \cong \mathbb{Z}^2.$$

Hence

$$H_1(K; \mathbb{Q}) \cong \mathbb{Q}^2.$$

This shows that rational homology counts the number of independent free loop generators.

Example 5: the real projective plane $\mathbb{RP}^2$

Let K be a triangulation of $\mathbb{RP}^2$. Then

$$H_1(K) \cong \mathbb{Z}/2, \quad H_2(K) = 0.$$

Since $\mathbb{Z}/2$ is torsion, it disappears after tensoring with $\mathbb{Q}$. Thus

$$H_1(K; \mathbb{Q}) = 0.$$

This is one of the most important examples: integral homology detects a nontrivial class, but rational homology does not detect it because it is torsion.

Example 6: a pure torsion group

Suppose a simplicial complex K has

$$H_i(K) \cong \mathbb{Z}/5.$$

Then the entire homology group is finite, so there is no free part. Therefore

$$H_i(K; \mathbb{Q}) \cong 0.$$

More generally, if

$$H_i(K) \cong F$$

for some finite abelian group F , then

$$H_i(K; \mathbb{Q}) = 0.$$

Thus purely torsion homology disappears completely over $\mathbb{Q}$.

Example 7: a mixed group $\mathbb{Z} \oplus \mathbb{Z}/2$

Suppose

$$H_i(K) \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

Then the free rank is 1, so

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}.$$

The $\mathbb{Z}$ -part survives, while the $\mathbb{Z}/2$ -part vanishes. This is exactly the meaning of the statement

$$\mathbb{Z} \oplus \mathbb{Z}/2 \rightsquigarrow \mathbb{Q}.$$

A standard topological example is the Klein bottle, whose first homology group is

$$H_1(\text{Klein bottle}) \cong \mathbb{Z} \oplus \mathbb{Z}/2.$$

Therefore

$$H_1(\text{Klein bottle}; \mathbb{Q}) \cong \mathbb{Q}.$$

Example 8: a mixed group $\mathbb{Z}^2 \oplus \mathbb{Z}/6$

Suppose

$$H_i(K) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/6.$$

Then the free rank is 2, so

$$H_i(K; \mathbb{Q}) \cong \mathbb{Q}^2.$$

Again, the torsion part disappears and only the free part remains.

Example 9: a contractible complex

If K is contractible, then

$$H_i(K) = 0 \quad (i \geq 1).$$

Therefore

$$H_i(K; \mathbb{Q}) = 0 \quad (i \geq 1).$$

So contractible spaces remain homologically trivial over $\mathbb{Q}$.

Why torsion disappears

The key algebraic reason is that if $x \in H_i(K)$ has finite order, say

$$mx = 0 \quad \text{for some } m > 0,$$

then in the tensor product $H_i(K) \otimes \mathbb{Q}$ we have

$$x \otimes q = mx \otimes \frac{q}{m} = 0 \otimes \frac{q}{m} = 0.$$

So every torsion element becomes zero after tensoring with $\mathbb{Q}$. That is why

$$\mathbb{Z}/n \otimes \mathbb{Q} = 0.$$

On the other hand,

$$\mathbb{Z} \otimes \mathbb{Q} \cong \mathbb{Q},$$

so each free $\mathbb{Z}$ -summand becomes one copy of $\mathbb{Q}$. More generally,

$$\mathbb{Z}^r \otimes \mathbb{Q} \cong \mathbb{Q}^r.$$

Summary table

$H_i(K)$	$H_i(K; \mathbb{Q})$	What happens
$\mathbb{Z}$	$\mathbb{Q}$	one free generator survives
$\mathbb{Z}^2$	$\mathbb{Q}^2$	two free generators survive
$\mathbb{Z}/2$	0	torsion disappears
$\mathbb{Z} \oplus \mathbb{Z}/2$	$\mathbb{Q}$	free part survives, torsion vanishes
$\mathbb{Z}^2 \oplus \mathbb{Z}/6$	$\mathbb{Q}^2$	same free rank, torsion vanishes
0	0	nothing remains

Conclusion

These examples show the general principle very clearly:

$$H_i(K) \cong \mathbb{Z}^r \oplus F \quad \implies \quad H_i(K; \mathbb{Q}) \cong \mathbb{Q}^r.$$

Thus rational homology remembers only the number of free generators and forgets all finite torsion.

9.3 Concrete picture of a triangulation of $\mathbb{RP}^2$ and why $H_1(\mathbb{RP}^2) = \mathbb{Z}/2$

A very concrete model of the real projective plane is

$$\mathbb{RP}^2 \cong D^2 / \{x \sim -x \text{ for } x \in \partial D^2\},$$

that is, a disk with opposite boundary points identified. This model is often the easiest starting point for understanding both a triangulation of $\mathbb{RP}^2$ and its first homology group.

Step 1: triangulate the disk

Take a closed disk and choose six equally spaced points on its boundary:

$$v_1, v_2, v_3, v_4, v_5, v_6,$$

with opposite points paired by

$$v_1 \sim v_4, \quad v_2 \sim v_5, \quad v_3 \sim v_6.$$

Add a center point c and triangulate the disk by joining c to each consecutive pair of boundary vertices. This gives six triangles:

$$\begin{aligned} T_1 &= \langle c, v_1, v_2 \rangle, & T_2 &= \langle c, v_2, v_3 \rangle, & T_3 &= \langle c, v_3, v_4 \rangle, \\ T_4 &= \langle c, v_4, v_5 \rangle, & T_5 &= \langle c, v_5, v_6 \rangle, & T_6 &= \langle c, v_6, v_1 \rangle. \end{aligned}$$

Before taking the quotient, this is just an ordinary fan of six triangles filling the disk.

Step 2: identify opposite boundary points

Now impose the antipodal identification on the boundary:

$$v_1 \sim v_4, \quad v_2 \sim v_5, \quad v_3 \sim v_6.$$

This also identifies opposite boundary edges:

$$\langle v_1, v_2 \rangle \sim \langle v_4, v_5 \rangle, \quad \langle v_2, v_3 \rangle \sim \langle v_5, v_6 \rangle, \quad \langle v_3, v_4 \rangle \sim \langle v_6, v_1 \rangle.$$

So the six triangles are paired:

$$T_1 \sim T_4, \quad T_2 \sim T_5, \quad T_3 \sim T_6.$$

If we denote the three boundary vertex classes by

$$a = [v_1 = v_4], \quad b = [v_2 = v_5], \quad d = [v_3 = v_6],$$

then after quotienting the six triangles become, schematically, three quotient triangles:

$$\langle c, a, b \rangle, \quad \langle c, b, d \rangle, \quad \langle c, d, a \rangle.$$

This gives a very concrete picture of $\mathbb{RP}^2$: it is built from triangular pieces, but with the outer boundary edges glued in pairs according to the antipodal rule.

Why this is often first viewed as a Δ -complex

After these identifications, some vertices and edges become identified with each other. Because of this, the quotient is most naturally described at first as a Δ -complex or CW-complex rather than as an honest simplicial complex in the strict sense. If one wants a genuine simplicial complex, one may subdivide further, for example by barycentric subdivision. Topologically, however, this quotient already gives the correct space $\mathbb{RP}^2$.

Step 3: the simpler CW-picture

For computing homology, the standard CW decomposition of $\mathbb{RP}^2$ is even simpler and more illuminating. It has:

- one 0-cell,
- one 1-cell a ,
- one 2-cell attached along the loop aa .

So the boundary of the 2-cell wraps twice around the 1-cell. In cellular notation this means

$$\partial e^2 = 2a.$$

Thus the cellular chain complex is

$$0 \longrightarrow C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \longrightarrow 0,$$

with

$$C_2 \cong \mathbb{Z}, \quad C_1 \cong \mathbb{Z}, \quad C_0 \cong \mathbb{Z},$$

and

$$\partial_2 = \times 2, \quad \partial_1 = 0.$$

So the chain complex becomes

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0.$$

Step 4: compute $H_1(\mathbb{RP}^2)$

By definition,

$$H_1(\mathbb{RP}^2) = \ker(\partial_1) / \text{im}(\partial_2).$$

Since $\partial_1 = 0$, we have

$$\ker(\partial_1) = \mathbb{Z}.$$

Since $\partial_2 = \times 2$, we have

$$\text{im}(\partial_2) = 2\mathbb{Z}.$$

Therefore

$$H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2.$$

So

$$\boxed{H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2.}$$

How this matches the triangle picture

In the disk model with opposite boundary points identified, the 2-dimensional part of the space is glued so that its boundary travels around the essential 1-dimensional loop twice. That is exactly what the CW description encodes by the attaching map

$$aa,$$

or equivalently by the boundary formula

$$\partial e^2 = 2a.$$

So the triangle picture and the CW picture describe the same topological fact: the fundamental 1-cycle becomes torsion of order 2.

Consequence for rational homology

Since

$$H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2$$

is purely torsion, it disappears after tensoring with $\mathbb{Q}$. Hence

$$H_1(\mathbb{RP}^2; \mathbb{Q}) = 0.$$

This is why $\mathbb{RP}^2$ is a basic example showing that rational homology detects only the free part of integral homology.

Summary

A concrete way to picture $\mathbb{RP}^2$ is:

- start with a disk,
- triangulate it by triangles around a center,

- identify opposite boundary points,
- observe that the resulting 2-cell wraps twice around the essential 1-cell.

This leads to the cellular chain complex

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

and therefore to

$$\boxed{H_1(\mathbb{RP}^2) = \mathbb{Z}/2.}$$

10 Triangulation and homology of real projective space

Statement

By describing a triangulation of S^n which is preserved under the antipodal map, show that $\mathbb{RP}^n$ has a triangulation. *[Be careful that the triangulation you describe actually comes from a simplicial complex! Some subdivision may be necessary.]*

Using the Mayer–Vietoris sequence, show that there is an exact sequence

$$0 \longrightarrow H_n(\mathbb{RP}^n) \longrightarrow \mathbb{Z} \longrightarrow H_{n-1}(\mathbb{RP}^{n-1}) \longrightarrow H_{n-1}(\mathbb{RP}^n) \longrightarrow 0$$

and that

$$H_i(\mathbb{RP}^{n-1}) \longrightarrow H_i(\mathbb{RP}^n)$$

is an isomorphism for $i < n - 1$. Hence show that

$$H_i(\mathbb{RP}^n) \cong \begin{cases} \mathbb{Z} & \text{if } i = 0 \text{ or if } i = n \text{ and } n \text{ is odd,} \\ \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i < n, \\ 0 & \text{otherwise.} \end{cases}$$

Deduce that $\mathbb{RP}^{2k}$ does not retract onto $\mathbb{RP}^{2k-1}$, and that any map

$$f : \mathbb{RP}^{2k} \rightarrow \mathbb{RP}^{2k}$$

has a fixed point.

Solution.

1. A triangulation of $\mathbb{RP}^n$.

Recall that

$$\mathbb{RP}^n = S^n / \{x \sim -x\}.$$

So to triangulate $\mathbb{RP}^n$, it is enough to choose a triangulation of S^n that is preserved by the antipodal map

$$a : S^n \rightarrow S^n, \quad a(x) = -x.$$

A standard simplicial model of S^n is the boundary of the $(n+1)$ -dimensional cross-polytope in $\mathbb{R}^{n+1}$. Its vertices are

$$\pm e_0, \pm e_1, \dots, \pm e_n,$$

where $e_0, \dots, e_n$ are the standard basis vectors. The boundary of this polytope is a simplicial complex homeomorphic to S^n . The antipodal map sends each vertex e_i to $-e_i$, and therefore sends simplices to simplices. Thus the antipodal map is simplicial with respect to this triangulation.

Taking the quotient by this free simplicial involution gives a finite Δ -complex structure on

$$S^n / \{\pm 1\} = \mathbb{RP}^n.$$

If one wants an honest simplicial complex in the strict sense, one may barycentrically subdivide. Hence $\mathbb{RP}^n$ has a triangulation.

Therefore

$\mathbb{RP}^n \text{ has a triangulation.}$

2. A Mayer–Vietoris decomposition of $\mathbb{RP}^n$.

Let $q : S^n \rightarrow \mathbb{RP}^n$ be the quotient map. We decompose $\mathbb{RP}^n$ into two subspaces U and V as follows. Let

$$U = q(\{x \in S^n : x_n \geq 1/3\} \cup \{x \in S^n : x_n \leq -1/3\}),$$

$$V = q(\{x \in S^n : |x_n| \leq 2/3\}).$$

Then $U \cup V = \mathbb{RP}^n$.

The space U is contractible: it comes from two antipodal caps in S^n , and after quotienting by the antipodal identification they become a single n -ball.

The space V deformation retracts onto the quotient of the equator

$$S^{n-1} / \{\pm 1\} = \mathbb{RP}^{n-1}.$$

Also, $U \cap V$ deformation retracts onto a copy of S^{n-1} . Thus

$$U \simeq *, \quad V \simeq \mathbb{RP}^{n-1}, \quad U \cap V \simeq S^{n-1}.$$

Now apply the Mayer–Vietoris sequence to the decomposition

$$\mathbb{RP}^n = U \cup V.$$

3. The Mayer–Vietoris sequence.

The Mayer–Vietoris sequence gives

$$\dots \rightarrow H_i(U \cap V) \rightarrow H_i(U) \oplus H_i(V) \rightarrow H_i(\mathbb{RP}^n) \rightarrow H_{i-1}(U \cap V) \rightarrow \dots$$

Since U is contractible and $V \simeq \mathbb{RP}^{n-1}$, this becomes

$$\dots \rightarrow H_i(S^{n-1}) \rightarrow H_i(\mathbb{RP}^{n-1}) \rightarrow H_i(\mathbb{RP}^n) \rightarrow H_{i-1}(S^{n-1}) \rightarrow \dots$$

If $i < n - 1$, then

$$H_i(S^{n-1}) = 0, \quad H_{i-1}(S^{n-1}) = 0,$$

so exactness gives an isomorphism

$$H_i(\mathbb{RP}^{n-1}) \xrightarrow{\cong} H_i(\mathbb{RP}^n).$$

Therefore

$$\boxed{H_i(\mathbb{RP}^{n-1}) \rightarrow H_i(\mathbb{RP}^n) \text{ is an isomorphism for } i < n - 1.}$$

For the top dimensions, the relevant piece of the sequence is

$$0 \rightarrow H_n(\mathbb{RP}^n) \rightarrow H_{n-1}(S^{n-1}) \rightarrow H_{n-1}(\mathbb{RP}^{n-1}) \rightarrow H_{n-1}(\mathbb{RP}^n) \rightarrow 0.$$

Since

$$H_{n-1}(S^{n-1}) \cong \mathbb{Z},$$

this becomes

$$\boxed{0 \rightarrow H_n(\mathbb{RP}^n) \rightarrow \mathbb{Z} \rightarrow H_{n-1}(\mathbb{RP}^{n-1}) \rightarrow H_{n-1}(\mathbb{RP}^n) \rightarrow 0.}$$

4. Understanding the map $\mathbb{Z} \rightarrow H_{n-1}(\mathbb{RP}^{n-1})$.

The map

$$H_{n-1}(S^{n-1}) \rightarrow H_{n-1}(\mathbb{RP}^{n-1})$$

is induced by the quotient map

$$p : S^{n-1} \rightarrow \mathbb{RP}^{n-1}.$$

Now there are two cases.

If $n - 1$ is even, then $\mathbb{RP}^{n-1}$ is non-orientable, hence

$$H_{n-1}(\mathbb{RP}^{n-1}) = 0.$$

If $n - 1$ is odd, then $\mathbb{RP}^{n-1}$ is orientable, and

$$H_{n-1}(\mathbb{RP}^{n-1}) \cong \mathbb{Z}.$$

In this case the quotient map p is a double covering, so on top homology the induced map is multiplication by 2:

$$p_* : \mathbb{Z} \rightarrow \mathbb{Z}, \quad p_* = \times 2.$$

Thus:

- if n is odd, then $H_{n-1}(\mathbb{RP}^{n-1}) = 0$;
- if n is even, then $H_{n-1}(\mathbb{RP}^{n-1}) \cong \mathbb{Z}$, and the map

$$\mathbb{Z} \rightarrow \mathbb{Z}$$

is multiplication by 2.

5. Inductive computation of the homology groups of $\mathbb{RP}^n$.

We now prove by induction on n that

$$H_i(\mathbb{RP}^n) \cong \begin{cases} \mathbb{Z} & \text{if } i = 0 \text{ or if } i = n \text{ and } n \text{ is odd,} \\ \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i < n, \\ 0 & \text{otherwise.} \end{cases}$$

Base case: $n = 0$. We have

$$\mathbb{RP}^0 = \{\text{point}\},$$

so

$$H_0(\mathbb{RP}^0) \cong \mathbb{Z}, \quad H_i(\mathbb{RP}^0) = 0 \quad (i > 0).$$

This matches the claimed formula.

Inductive step. Assume the formula holds for $\mathbb{RP}^{n-1}$.

For every $i < n - 1$, we already know from Mayer–Vietoris that

$$H_i(\mathbb{RP}^n) \cong H_i(\mathbb{RP}^{n-1}),$$

so all lower homology groups are inherited unchanged.

It remains to determine $H_n(\mathbb{RP}^n)$ and $H_{n-1}(\mathbb{RP}^n)$.

Case 1: n odd.

Then $n - 1$ is even, so by the inductive hypothesis

$$H_{n-1}(\mathbb{RP}^{n-1}) = 0.$$

Therefore the exact sequence

$$0 \rightarrow H_n(\mathbb{RP}^n) \rightarrow \mathbb{Z} \rightarrow H_{n-1}(\mathbb{RP}^{n-1}) \rightarrow H_{n-1}(\mathbb{RP}^n) \rightarrow 0$$

reduces to

$$0 \rightarrow H_n(\mathbb{RP}^n) \rightarrow \mathbb{Z} \rightarrow 0 \rightarrow H_{n-1}(\mathbb{RP}^n) \rightarrow 0.$$

So

$$H_n(\mathbb{RP}^n) \cong \mathbb{Z}, \quad H_{n-1}(\mathbb{RP}^n) = 0.$$

Case 2: n even.

Then $n - 1$ is odd, so

$$H_{n-1}(\mathbb{RP}^{n-1}) \cong \mathbb{Z}.$$

Also the map $\mathbb{Z} \rightarrow \mathbb{Z}$ is multiplication by 2. Thus the exact sequence becomes

$$0 \rightarrow H_n(\mathbb{RP}^n) \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow H_{n-1}(\mathbb{RP}^n) \rightarrow 0.$$

Since multiplication by 2 is injective,

$$H_n(\mathbb{RP}^n) = 0,$$

and since its cokernel is $\mathbb{Z}/2$,

$$H_{n-1}(\mathbb{RP}^n) \cong \mathbb{Z}/2.$$

Combining these two cases with the lower-dimensional isomorphisms gives the full formula:

$$H_i(\mathbb{RP}^n) \cong \begin{cases} \mathbb{Z} & \text{if } i = 0 \text{ or if } i = n \text{ and } n \text{ is odd,} \\ \mathbb{Z}/2 & \text{if } i \text{ is odd and } 0 < i < n, \\ 0 & \text{otherwise.} \end{cases}$$

6. Deduction: $\mathbb{RP}^{2k}$ does not retract onto $\mathbb{RP}^{2k-1}$.

Suppose there were a retraction

$$r : \mathbb{RP}^{2k} \rightarrow \mathbb{RP}^{2k-1}.$$

Let

$$j : \mathbb{RP}^{2k-1} \hookrightarrow \mathbb{RP}^{2k}$$

be the inclusion. Then

$$r \circ j = \text{id}_{\mathbb{RP}^{2k-1}}.$$

Passing to homology in degree $2k-1$, we get

$$r_* \circ j_* = \text{id} \quad \text{on} \quad H_{2k-1}(\mathbb{RP}^{2k-1}) \cong \mathbb{Z}.$$

But

$$H_{2k-1}(\mathbb{RP}^{2k}) \cong \mathbb{Z}/2.$$

So j_* is a homomorphism

$$j_* : \mathbb{Z} \rightarrow \mathbb{Z}/2.$$

No homomorphism

$$r_* : \mathbb{Z}/2 \rightarrow \mathbb{Z}$$

can satisfy

$$r_* \circ j_* = \text{id}_{\mathbb{Z}},$$

because the identity on $\mathbb{Z}$ cannot factor through the finite group $\mathbb{Z}/2$. This is impossible.

Hence there is no retraction. Therefore

$$\boxed{\mathbb{RP}^{2k} \text{ does not retract onto } \mathbb{RP}^{2k-1}.}$$

7. Deduction: every map $f : \mathbb{RP}^{2k} \rightarrow \mathbb{RP}^{2k}$ has a fixed point.

Now consider a continuous map

$$f : \mathbb{RP}^{2k} \rightarrow \mathbb{RP}^{2k}.$$

We use the Lefschetz fixed point theorem.

From the homology computation above, for even-dimensional projective space $\mathbb{RP}^{2k}$, all positive-dimensional homology groups are either zero or finite torsion. Therefore with rational coefficients,

$$H_0(\mathbb{RP}^{2k}; \mathbb{Q}) \cong \mathbb{Q}, \quad H_i(\mathbb{RP}^{2k}; \mathbb{Q}) = 0 \quad (i > 0).$$

Hence the Lefschetz number is

$$L(f) = \sum_i (-1)^i \operatorname{tr} (f_* : H_i(\mathbb{RP}^{2k}; \mathbb{Q}) \rightarrow H_i(\mathbb{RP}^{2k}; \mathbb{Q})).$$

Since only H_0 contributes, and f_* on H_0 is always the identity, we get

$$L(f) = 1.$$

Since $L(f) \neq 0$, the Lefschetz fixed point theorem implies that f has a fixed point.

Therefore

Every map $f : \mathbb{RP}^{2k} \rightarrow \mathbb{RP}^{2k}$ has a fixed point.